

Client requires website with following functionalities:

I. Website Admin will make Plans of certain tenures like 6 months, 1 year, 2 years, 3 years etc. that consists of benefits provided and the gain attained upon maturity from which registered user can opt any one, select the payment amount contribution (either monthly or Lum sum payment) and make payment through payment gateway integration.

II. When user select the plan and payment amount, then he would be able to see:

- Invested or Contributed amount.
- Amount which can be redeemable after the completion of selected Plan tenure and thirdly the amount which user can redeem if they do any educational course from the institute or choose any other things from the list provided by the institute. Amount which user can redeem will be calculated according to annual/monthly interest defined by the web admin for that plan.
- Amount which is shown to user for opting any course/Item provided by institute, it would be again calculated according to annual/monthly interest defined by the web admin for that plan.

III. **Plan Discontinuation:** When an Admin discontinues a plan, existing users enrolled in that plan shall continue to receive the value applicable at maturity.

- User cannot redeem the money back before the maturity of the selected plan tenure even if they stop paying the scheduled instalment irrespective of whether the plan gets discontinued by the admin.
- **Discontinued Plan Availability:** Once a Plan is discontinued, it shall no longer be available for display, selection, or enrolment by new users.
- **Existing Enrolled Users:** Discontinuation of a Plan shall not terminate or discontinue the Plan for users who were already enrolled before the discontinuation. Such users shall continue under the applicable Plan terms until the Plan reaches its maturity date.
- **Discontinued Plan Maturity:** For an existing enrolled user, the discontinued Plan shall remain active until its configured maturity date. Upon reaching the maturity date, the Plan shall transition to MATURED and become eligible for the normal redemption flow.
- **Discontinuation Financial Treatment:** No haircut, deduction, or penalty shall be applied solely because the Plan was discontinued. The applicable plan value shall be calculated and settled according to the applicable Plan terms, including any applicable pro-rated value. Here pro-rated value is used for the users who were unable to submit all the instalments till the maturity period. So, whatever amount had been paid by the users, the final settlement will be done on the basis of the amount deposited by the user.
- Plan cancellation isn't required in this project.
- When an Admin discontinues a Plan, the Plan shall move to DISCONTINUED status for existing enrolled users. The Plan shall remain DISCONTINUED until its configured maturity date. Existing enrolled users shall continue under the applicable Plan terms, and upon maturity the Plan shall transition from DISCONTINUED → MATURED.
- A DISCONTINUED Plan shall not be available for new user enrolment.

IV. Final redeemable amount will be calculated according to the amount in the user account at the time of plan maturity.

V. User can redeem partial amount after the plan maturity and can re-invest pending amount by choosing any new plan. There is no limit to partial redemptions.

VI. Autopayment receipt is generated and sent to user email when their payment is deducted. An option to download will be available to download the receipts, reports and transactions.

VII. Option to deduct auto payment from user attached credit/debit card, UPI, Net banking or digital wallet as per the payment frequency selected by the user while opting the plan.

VIII. **Referral Commission Records and Views**

The system shall maintain commission transaction records within the **Unified Financial Ledger** for each Referrer to record commission transactions generated from eligible payments received from their directly Referred Users, irrespective of the number of plans opted for by each Referred User. The Commission Records shall provide the following views:

1. **Per-Referred-User Commission View**

- Display commission details for each individual referred user.
- Show commission earned from all plans associated with the referred user.
- Include commission generated from both **active and expired plans**.

- Display **upcoming commission** expected from currently active plans based on eligible future payments.

2. Referrer-Level Commission Summary

- Provide an aggregated view of commission across all referred users.
- Display total commission earned, including commission generated from active and expired plans.
- Display the total upcoming commission associated with currently active plans.

3. Plan-Level Commission Details

- For each referred user, the ledger shall identify the plans against which commission was earned or is expected.
- Commission shall be calculated and recorded based on the **actual payment amount received** from the referred user and the applicable commission rules.

4. Commission Transaction Records: Commission transactions shall be recorded in the unified financial ledger with the applicable commission reference, Referrer User ID, Referred User ID, commission amount, status and transaction timestamp.

5. Commission Records/View: The system shall provide a consolidated financial view of a Referrer's commission at both the **individual referred-user level** and the **overall referrer level**, while maintaining the commission history and upcoming commission associated with each plan. These records shall be derived from the unified financial ledger.

IX. **Commission Frequency:** If the Admin configures a Referrer for **Recurring Commission**, the Referrer shall continue to receive commission for every four consecutive successful payments received from the directly Referred User throughout the tenure of the applicable plan. No commission shall be generated for failed or unsuccessful payments. Failed payments shall not terminate the Referrer's recurring commission eligibility; commission eligibility shall remain applicable for the remaining tenure of the plan and shall cease when the plan reaches its maturity/expiry. If the Admin configures a Referrer for **One-Time Commission**, the Referrer shall receive commission only once in accordance with the applicable commission rules, regardless of subsequent eligible payments.

X. **Commission Crediting:** Eligible commission shall be credited to the Referrer's redeemable balance through the Unified Financial Ledger.

XI. Referrer will be able to view a record of the referred users and their respective commission received.
Eligible Commission → Commission Transaction → Unified Financial Ledger → Commission Credited → Redeemable Balance → Available for Redemption

XII. **Referral Code and Relationship Creation:** Each registered user shall have one active unique referral code that may be shared with prospective users. A user may regenerate/rotate their referral code when required. Upon regeneration, the previous code shall become inactive for new registrations and the newly generated code shall become the user's active referral code. Existing referral relationships shall remain associated with the original referring user and shall not be affected by code regeneration. A referral relationship shall be created only when a new user uses an existing user's referral code during registration and successfully completes the applicable registration requirements. The user whose referral code is used shall become the Referrer, and the newly registered user shall become the Referred User.

New User Registration → Uses Referral Code → Registration Successfully Completed → Referral Relationship Created

XIII. If student redeems balance for course, then certain reward points will be available to the student for its next course which cannot be redeemed in any other manner. If reward points are available, then the course fees will be adjusted accordingly. Reward points cannot be withdrawn.

Reward Percentage Calculation: The reward percentage shall be calculated on the **course fee amount applicable to the approved course redemption**. The reward points/amount shall be calculated as the applicable course fee multiplied by the configured reward percentage. The course fee shall be the authoritative calculation basis for course redemption rewards. The Reward Percentage used for calculation shall be the Reward Percentage configured for the applicable plan, or the common Reward Percentage configured by Admin where the system is using a common reward configuration.

XIV. Plans when opted, a view will appear with plan detail as set by the admin and inputs that includes the list of preset amounts and list of payment frequency set by the admin. These inputs of amount and payment frequency will be used for auto pay feature. Plan created by user talks about the benefits and gains during until the plan maturity whereas the amount selected by the user is based on the minimum investment required for the plan and the payment frequency. When user submit the plan, then auto-debit will be effective upon auto-payment is setup by the user for the plan. Earlier the selected duration is the frequency of Autopay that will take place until the maturity of the plan. User can have multiple plans available to opt

and will be active as per the plan maturity period. So basically, a plan in this project means a screen where plan detail set by admin are available along with the inputs on the basis of which Autopay will work until the plan reaches the maturity.

XV. Direct Referral Commission: The Referrer shall be eligible to receive commission only from eligible payments made by users directly referred by that Referrer. Commission shall be calculated independently for each directly referred user and their eligible plans. A Referrer shall not receive commission from any user referred by their directly Referred User or from any other downstream or indirect referral relationship.

XVI. If admin changes the plan details like interest, etc then existing plans shouldn't be affected by the change. If a Plan is discontinued, the existing enrolled Plan shall remain DISCONTINUED until its originally configured maturity date and shall transition to MATURED only upon reaching that maturity date.

XVII. User with get notifications and messages regarding grace period provided for failed payments. This grace period is set by admin. If payment is received within the grace period the referrer will receive commission otherwise for skipped payments no commission will be generated. In short for every four consecutive successful payments commission will be generated as long as user have active plan(s) if referrer is allowed to receive recurring commission by the admin. The late payments won't attract any penalties. Commission already withdrawn by the referrer shall likewise never be returned or recovered.

XVIII. Single-Level Referral Relationship: The system shall maintain only direct, single-level Referrer → Referred User relationships. A Referrer may refer one or more users by having each new user use the Referrer's referral code during registration. A Referred User may subsequently become a Referrer and refer one or more newly registering users using their own referral code. Each subsequent referral shall be treated as an independent direct referral relationship and shall not create any multi-level, hierarchical, or indirect commission relationship. Let's say, $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow D$. It does **not** mean: $A \rightarrow B \rightarrow D$, which is a multi-level commission relationship. Instead, there are two independent relationships: $A \rightarrow B$, $B \rightarrow D$.

XIX. Referral Validity Period – The Admin shall configure a positive referral validity period in whole days. The system shall validate the configured value against the supported minimum and maximum range. Each referral relationship shall receive an expiry date calculated from its creation date/time using the validity period effective at the time of relationship creation. Referral validity can be applicable to either selected referrers or all referrers.

Proposed default 30–1095 days (min 30 to prevent instant expiry, max 3 years).

XX. Referral Commission Discontinuation – Cancellation shall prevent **future referral commission accrual** associated with the referrer from the effective cancellation time. An authorized administrator may cancel an active referral relationship for a user stating a reason for cancellation as written by the admin in cancellation comments.

When a referral is cancelled:

- No new referral commission shall accrue from the cancellation date/time.
- Any commission that was already accrued before the cancellation shall continue through the normal commission approval process.
- Existing commission shall not be marked CANCELLED solely because the referral relationship was subsequently cancelled.
- The cancellation reason, effective date/time, and Admin details shall be recorded in the audit log.
- Refer to business rule XXI for the non-reversal treatment of approved commission.

XXI. Referral commission becomes non-reversible once it reaches APPROVED status. Approved, credited, available, or withdrawn commission shall not be clawed back, reversed, cancelled, or reduced due to subsequent referral cancellation, Referral Validity Period, payment failure, plan discontinuation, or other subsequent events. Commission already withdrawn by the referrer shall not be returned or recovered under any circumstances.

XXII. Redemption Shortfall Resolution: If the requested redemption amount exceeds the user's **Available Margin**, the redemption request shall be placed in **AWAITING_SHORTFALL_RESOLUTION** status and the system shall calculate and display the **Shortfall Amount**.

The **entire redemption request shall remain financially unprocessed** until the shortfall is resolved outside the system, the user submits the required resolution details/evidence, and the Admin verifies and approves the resolution.

No partial or full redemption settlement, final redemption ledger debit, or reduction of the Redeemable Balance attributable to the redemption shall occur while the request is awaiting shortfall resolution or Admin approval.

The system shall **not process the Available Margin portion separately** from the Shortfall Amount. A redemption request having a shortfall shall be treated as **one redemption request for financial processing purposes**.

Resolution of the shortfall outside the system shall **not itself create a redemption ledger debit**. Any external/offline amount used to resolve the shortfall shall be recorded as shortfall-resolution information/evidence and shall not be treated as a deduction from the user's Redeemable Balance unless a separate, explicitly defined financial transaction applies.

The user must either:

1. **Resolve the shortfall outside the system** and submit the required resolution details/evidence for Admin verification; or
2. **Cancel the redemption request.**

If the user cancels the request, the redemption shall be marked **CANCELLED** and any associated reservation shall be released.

After successful Admin approval, the redemption shall become eligible for financial processing. The system shall then process the redemption in accordance with the applicable settlement mechanism and shall create the corresponding financial ledger transaction(s).

The resulting ledger transaction(s) shall be linked to the relevant **Redemption Request ID** and shall accurately represent the amount and nature of the actual financial transaction. The system shall not create a ledger debit merely to represent the offline shortfall unless such debit represents an actual, separately defined financial transaction.

Shortfall Amount = Requested Redemption Amount – Available Margin

Submission vs Financial Processing Clarification:

The existence of a Shortfall shall not prevent the user from submitting the redemption request.

A redemption request may be submitted even when:

Requested Redemption Amount > Available Margin

In such a case, the system shall accept the request and place it in:

AWAITING_SHORTFALL_RESOLUTION

The Shortfall shall prevent **approval and financial processing**, but shall not prevent **request submission**.

No partial redemption of the Available Margin portion shall occur while the Shortfall remains unresolved.

XXIII. No Financial Impact Before Approval: A redemption request with a shortfall shall **not reduce the user's Redeemable Balance or create a final ledger debit** while it is in

AWAITING_SHORTFALL_RESOLUTION or awaiting Admin approval.

Any reservation associated with the request shall remain **separately tracked from financial transactions** and shall not constitute a ledger debit or financial settlement.

If the request is cancelled, any associated reservation shall be released in accordance with the applicable reservation rules.

XXIV. Admin Approval: Only after the Admin verifies that the shortfall has been resolved may the redemption move to **APPROVED** and proceed to financial processing.

A redemption request with a shortfall shall be treated as a **single redemption request for financial processing**. The system shall **not financially process or create a ledger debit for the Available Margin portion separately** while the shortfall remains unresolved or awaiting Admin approval.

After Admin approval, the system shall proceed with financial processing in accordance with the applicable redemption and settlement rules. The resulting ledger transaction(s) shall represent only the **actual financial transaction(s)** that occur as part of the approved redemption.

Shortfall → Offline Resolution → Admin Verification → Approval → Financial Processing → Ledger Impact → Balance Update

XXV. **Available Margin**

Available Margin already reflects amounts reserved by pending redemptions; therefore, a pending redemption amount must not be deducted from Available Margin again when calculating the shortfall.

Available Margin Formula:

Available Margin = Actual Redeemable Balance – Total Active Reserved/Committed Redemption Amount

The system shall calculate a user's Available Margin as the portion of the Actual Redeemable Balance that is not committed to existing pending redemption requests or other active reservations.

The system shall consider all applicable active reservations across redemption categories when calculating Available Margin.

An active reservation may reduce Available Margin but shall **not constitute a financial ledger debit or a separate reduction of the Actual Redeemable Balance**.

A redemption ledger debit shall occur only when the redemption is financially processed in accordance with the applicable redemption and settlement rules.

Interest Calculation Reference

For any ledger entry generated by an interest calculation, the entry shall include the **Interest Calculation Method ID and configuration/version ID** used for that calculation.

XXVI. Interest Calculation Formula Configuration:

a) The Admin shall configure and manage an **Interest Calculation Method** by selecting one of the following formula types and specifying its required parameters:

- **Simple Interest**
- **Compound Interest**, with configurable compounding frequency
- **Custom Parameterized Formula**

b) For Simple Interest and Compound Interest, the Admin shall configure the applicable **interest rate, tenure basis, and interest period**, including predefined periods such as **weekly, monthly, quarterly, and annually**, with a custom period option where permitted. For Compound Interest, the Admin shall additionally specify the **compounding frequency**.

c) A Custom Parameterized Formula shall use only the following system-approved variables and predefined mathematical operators:

- Principal
- Rate
- Tenure
- Elapsed Days

Custom Formula Validation: Custom formulas shall use only the approved variables Principal, Rate, Tenure, and Elapsed Days, together with predefined mathematical operators supported by the system. The system shall validate syntax, operator usage, nesting depth, expression length, unsupported variables/operators, division by zero and non-finite results before saving or applying the formula. Invalid formulas shall be rejected.

- +
- -
- *
- /
- parentheses
- maximum expression length
- maximum nesting depth
- division-by-zero → reject
- NaN / Infinity → reject
- negative result → reject

d) The Admin shall be able to apply an Interest Calculation Method to selected plans, including newly created or existing plans. Once applied, the method and its parameters shall be **snapshotted and locked for that plan**. The snapshot shall include the formula type, formula/parameters, interest rate, tenure/interest-period basis, compounding frequency where applicable, and configuration version.

e) Any subsequent modification to an Interest Calculation Method shall apply only to plans created or assigned after the modification and shall **not retroactively change or recalculate interest already accrued** under an existing plan.

f) Every interest calculation shall be recorded in the **ledger** with a reference to the exact Interest Calculation Method and configuration/parameter version used, enabling **audit and reproducibility**.

Custom Formula Constraints: Maximum expression length shall be 100 characters and maximum nesting depth shall be 5.

XXVII. Failed Payment and Retry Management:

- The system shall support automatic retry of failed recurring Plan payments through the configured payment gateway. The Admin shall be able to configure the maximum number of retry attempts, retry interval, and applicable grace period.
- When a scheduled Plan payment fails, the system shall record the failed payment attempt and notify the user. The system shall automatically retry the payment according to the configured retry policy until the payment succeeds, the maximum retry attempts are exhausted, or the configured grace period expires, whichever occurs first.

- Failure to successfully complete a scheduled payment, including exhaustion of all configured retries and expiry of the applicable grace period, **shall not result in cancellation or suspension of the enrolled Plan.**
- The system shall continue to maintain the Plan until its applicable maturity date. At maturity, the system shall calculate the amount payable to the user based on the **actual amount and applicable duration for which payments were successfully received**, using the defined pro-rata maturity calculation. The resulting amount shall be recorded in the ledger and made payable to the user through the applicable redemption/maturity process.
- All successful payments, failed attempts, retries, and applicable adjustments to the maturity calculation shall be recorded in the ledger for auditability.
- **Grace Period:** The Admin shall configure the payment grace period in hours. The value shall be a positive whole number within the configured system-supported range. The configured grace period shall start from the scheduled payment due date/time and shall determine the period during which a failed payment may be successfully recovered. **24 hours default**, allow 0–168h (up to 7 days) configurable.
- **Retry Configuration:** The Admin shall configure the maximum number of automatic retry attempts and the retry interval. The system shall retry only failures identified as retrievable by the payment gateway. Non-retrievable or terminal failures shall not be retried automatically. Retry attempts shall stop when payment succeeds, the maximum retry count is reached, the grace period expires, or the gateway identifies the failure as non-retrievable. User will then get an option to pay using other modes of payments like UPI, Credit/Debit cards, Net banking, etc, which are manual payment modes. These modes will appear when retry limit and grace period had reached their limits. Retry intervals shall follow a configurable retry schedule; where no schedule is configured, a fixed configured interval shall be used. Retry count/interval — default 3 retries, interval 24h between attempts (so a 3-retry cycle spans ~3 days), configurable range 1–5 retries / 1–72h interval.
- **Manual Payment Fallback:** After the configured retry attempts are exhausted or the grace period expires, the system shall provide the user with the available manual payment options. The Admin shall configure a manual payment window during which the user may complete the missed scheduled payment manually. The default manual payment window shall be 24 hours, with a configurable range of 1 to 72 hours. The manual payment window shall be measured from the point at which automatic recovery ends. The manual payment window shall apply only after automatic recovery ends and shall not extend or modify the configured retry count or grace period. A manual payment successfully received within this configured window shall be treated as a successful payment for the corresponding scheduled instalment and shall be recorded in the Unified Financial Ledger. A manual payment received after the configured manual payment window shall not be treated as a successful payment for that scheduled instalment.

XXVIII. **Concurrent reservations across categories:**

Concurrent Redemption Reservations: The system shall maintain a consolidated reservation amount across all active, pending redemption requests, irrespective of redemption category. When calculating Available Margin for a new redemption request, the system shall include the reserved amounts of all other active redemption requests for the same user.

A new redemption request shall be evaluated against the remaining Available Margin after accounting for all existing active reservations. The system shall not allow concurrent redemption requests to collectively exceed the user's Actual Redeemable Balance.

Example:

Actual Redeemable Balance = ₹10,000

Pending Course Reservation = ₹4,000

Pending Gadget Reservation = ₹3,000

Pending Franchisee Reservation = ₹2,000

Total Reserved = ₹9,000

Available Margin = ₹10,000 – ₹9,000 = ₹1,000

SUBMITTED → ADMIN REVIEW → AWAITING_SHORTFALL_RESOLUTION

XXIX. **Redemption Reservation and Available Margin:** The system shall maintain a reservation against each active redemption request that has committed or reserved an amount against the user's Actual Redeemable Balance. Available Margin shall be calculated as:

Available Margin = Actual Redeemable Balance – Total Active Reserved Redemption Amount

The Total Active Reserved Redemption Amount shall include reservations from all active redemption requests across all redemption categories for the same user. The system shall recalculate Available Margin whenever a reservation is created, modified, approved, rejected, cancelled, or released. A new redemption request shall not be permitted to cause the combined active reservations to exceed the Actual Redeemable Balance.

XXX. Payment Method: The user may change the payment method associated with an ACTIVE or DISCONTINUED Plan, subject to applicable payment gateway validation and authorization requirements. Changing the payment method shall not modify the Plan's payment amount, payment frequency, principal already paid, or accrued interest. For a DISCONTINUED Plan, the payment method may be changed only for scheduled payments that remain due before the Plan's maturity date.

XXXI. Unified Ledger Model: The system shall maintain a single unified financial ledger for all applicable financial transactions, including plan transactions, interest transactions, commission transactions and redemption transactions. Commission transactions shall be recorded in the unified ledger and shall contribute to the user's applicable redeemable/available balance in accordance with the commission rules. A separate Commission Ledger shall not be maintained as an independent financial ledger.

XXXII. Commission Calculation: The Admin shall configure the commission type for each Referrer as either **Recurring Commission** or **One-Time Commission**, along with the applicable commission percentage.

a) Recurring Commission: Recurring Commission shall be generated after every **four consecutive successful payments** received under the plan. Once four consecutive successful payments have been received, the commission cycle shall restart and the next four consecutive successful payments shall become eligible for commission. This cycle shall continue repeatedly until the **Plan Maturity Date**. Failed payments shall not count toward the four-payment cycle.

b) Failed Payment: A failed payment shall not generate commission and shall not count as a successful payment toward the four-payment commission limit.

c) Late Payment within Grace Period: If a scheduled payment initially fails but is subsequently received successfully within the configured grace period, the payment shall be treated as a successful payment for commission calculation, and commission shall be calculated on the successfully received amount.

d) Failed Payment after Grace Period: If the payment is not successfully received within the applicable grace period, no commission shall be generated for that payment. Future eligible successful payments shall remain subject to the applicable commission rules.

e) One-Time Commission: If the Admin configures the Referrer for One-Time Commission, the Referrer shall receive commission at the configured percentage on the **first successful payment** made by each directly Referred User under the applicable plan. No further commission shall be generated for subsequent payments under that plan.

f) Commission Calculation Basis: Commission shall be calculated on the **actual payment amount successfully received** from the directly Referred User.

g) Commission Payout Timing: The commission payout shall be processed according to the payout frequency configured by the Admin (weekly, monthly, quarterly or yearly). The default payout frequency shall be monthly unless the Admin configures a different frequency. Payout frequency shall not change the commission calculation or eligibility rules.

XXXIII. Percentage Configuration and Rounding: Interest, reward and commission percentages shall be configured as decimal percentages **from 0.00% to 100.00%, inclusive**, with up to two decimal places. Values outside this range shall be rejected by the system. Monetary calculations shall be rounded to two decimal places using the **Round Half Up** method. Where the third decimal place is 5 or greater, the second decimal place shall be increased by one; where it is less than 5, the second decimal place shall remain unchanged.

For ₹:

- Percentage: **up to 2 decimal places**
- Money: **2 decimal places**

Percentage Range: Interest, Reward and Commission percentages shall be configured between **0.00% and 100.00%, inclusive**. Values below 0.00% or above 100.00% shall not be accepted by the system. The percentage configuration shall support up to two decimal places.

Validation Examples:

- 0.00% → Allowed
- 10.00% → Allowed
- 100.00% → Allowed

- 100.01% → Rejected
- Negative percentage → Rejected

XXXIV. Money Mechanics:

- **Currency and Monetary Precision:** All monetary amounts shall be denominated in **Indian Rupees (INR)**. Monetary values shall be stored and calculated with a precision of two decimal places (paise). All monetary calculations shall be rounded to two decimal places using the **Round Half Up** method defined in Rule XXXIII. The same rounding method shall apply consistently to interest, commission, redemption, balance, payment, and ledger calculations.
- **Interest Day-Count Convention:** Where interest or pro-rata maturity calculations are based on elapsed days, the system shall use the **Actual/365 day-count convention**, unless a plan-specific calculation method explicitly defines another supported convention.
- **Interest Calculation Rounding and Compounding:** Interest shall be calculated using the applicable Interest Calculation Method and configured compounding frequency. For compound interest, the calculated interest shall be added to the applicable principal/base amount at each configured compounding interval. Intermediate calculations shall retain system-supported precision, and the final interest amount credited or recorded in the ledger shall be rounded to two decimal places using the Round Half Up method defined in Rule XXXIII. No additional rounding shall be applied unless specifically defined by the selected Interest Calculation Method.
- **Offline Shortfall Verification:** For an offline shortfall payment, the Admin shall record the payment reference/receipt number, payment date, amount, payment method and verification comments before marking the shortfall as verified. Supporting proof may be uploaded where required. An offline shortfall shall not be treated as resolved until verified by an authorized Admin. If a previously verified payment is subsequently determined to be invalid, the system shall record the reversal/exception through an audit-controlled adjustment process.
- **Donation Processing:** The Admin shall configure or select the eligible donation recipient/entity for each donation request. Approved donations shall be processed by Admin offline after obtaining the user's consent. Admin approval shall trigger the applicable financial processing through the Unified Financial Ledger, and the approved donation amount shall be debited from the user's Redeemable Balance. The Available Margin shall be recalculated after the balance update. The system shall record the recipient, approved donation amount, transaction reference, processing status, consent/approval details and relevant dates. Where required, the applicable donation receipt/document shall be generated or recorded.
- This Offline Shortfall Verification procedure is the ****system-wide verification procedure**** and shall apply to all redemption categories involving an offline shortfall, including Course, Gadgets & Accessories, and Franchisee redemption.
- **Currency Display Convention:** All monetary amounts shown anywhere in the system, including plans, payments, interest, commission, redemption amounts, balances, ledger entries, dashboards, notifications, reports, and examples, shall be displayed with the INR currency symbol (₹) or the explicit INR designation. Monetary values shall not be displayed as standalone numbers where the context could be ambiguous.

XXXV. Concurrency / race conditions:

Concurrent Redemption Control: Available Margin validation and reservation creation shall be performed atomically. The system shall prevent two concurrent redemption requests from reserving the same available balance. Appropriate database transaction isolation, row-level locking or optimistic locking shall be used to ensure that the combined active reservations never exceed the Actual Redeemable Balance.

Inventory Reservation: When a gadget/accessory redemption request is submitted, the system shall reserve the required inventory quantity in addition to the financial reservation. Reserved inventory shall not be available for another redemption request. The reservation shall be released when the request is cancelled/rejected/expired and converted to a stock deduction when the request is approved.

Redemption Request Expiry: The Admin shall configure the maximum period for which a redemption request may remain in PENDING or AWAITING_SHORTFALL_RESOLUTION.

- Default expiry period: 7 calendar days.
- Configurable range: 1 to 30 calendar days.
- The expiry period shall be measured from the time the request enters the applicable status.
- When the configured period expires without the required action, the request shall automatically move to EXPIRED status and all associated financial and inventory reservations shall be released.

- The expiry configuration shall apply uniformly to all applicable redemption categories, including Course, Gadgets & Accessories, and Franchisee redemption.
- Any change to the expiry configuration shall apply to requests created or entering the applicable status after the configuration change and shall not retroactively alter an already-running expiry period.

Redemption Request Status:

- PENDING
- AWAITING_SHORTFALL_RESOLUTION
- APPROVED
- REJECTED
- CANCELLED
- EXPIRED

Plan Status:

- ACTIVE
- MATURED
- DISCONTINUED
- PARTIALLY_REDEEMED
- REDEEMED

XXXVI. Duplicate Payment Handling and Idempotency

The system shall prevent the same payment from being credited, recorded, or processed more than once, including where duplicate gateway callbacks/webhooks, user retries, network retries, scheduled-payment retries, or delayed gateway responses are received.

1. Idempotency Key

Every payment initiation request shall have a unique **Idempotency Key** generated by the backend. The Idempotency Key shall be associated with the intended payment transaction and shall remain unchanged when the same payment request is retried.

The system shall persist the Idempotency Key and shall enforce uniqueness at the database level.

If the same Idempotency Key is received again:

- the system shall not create a new payment transaction;
- the system shall not create another investment/ledger entry;
- the system shall not generate another commission;
- the system shall not send another successful-payment business event;
- the system shall return/reference the result of the original transaction.

2. Gateway Transaction Reference

For payments processed through Razorpay, the system shall store the gateway-provided unique payment identifiers, including the applicable Order ID, Payment ID, and other relevant gateway reference/receipt identifiers.

A successful gateway transaction identifier shall be unique within the payment records.

A webhook/callback for an already processed gateway Payment ID shall be treated as a duplicate event and shall not result in another financial transaction.

3. Definition of a Duplicate Payment

A payment shall be considered a duplicate when any of the following is true:

- a. The same Idempotency Key has already been processed.
- b. The same gateway Payment ID has already been successfully processed.
- c. The same gateway Order ID is associated with a payment that has already been successfully settled and the incoming event attempts to create another settlement for that same payment.
- d. The same payment event/webhook is delivered more than once by the payment gateway.
- e. The client retries the same payment operation after a timeout where the original payment request has already succeeded.
- f. An automatic retry and a previous payment attempt result in the same underlying gateway payment being reported as successful more than once.

4. Business Transaction Uniqueness

A successful payment shall result in exactly one applicable financial posting for the corresponding payment event.

For each successful payment, the system shall create the required unified ledger transaction only once. The payment record and ledger transaction shall be linked using a unique internal Payment Transaction ID. No duplicate payment shall:

- increase the user's invested/contributed amount twice;
- increase redeemable or plan balances twice;
- trigger interest-related processing twice;
- generate referral commission twice;
- count twice toward the recurring commission cycle;
- generate duplicate receipts or duplicate payment-success business events.

Successful payments and related transactions must be recorded in the unified ledger and that commission is calculated from successfully received payments.

5. Razorpay Webhook Signature Verification

All Razorpay webhook requests shall be authenticated before any webhook business processing is performed.

The system shall verify the Razorpay webhook signature using the configured Razorpay webhook secret and the exact raw webhook request payload received from Razorpay.

If signature verification fails, the system shall:

- reject the webhook request;
- not create or update any financial transaction;
- not update the user's balance;
- not generate commission;
- record the failed verification event in the audit log.

Signature verification shall be completed before duplicate detection or any other financial/business processing of the webhook.

The webhook secret shall be stored securely and shall not be exposed in application logs, API responses, or user-facing interfaces.

6. Webhook Idempotency

Razorpay webhook processing shall be idempotent.

The system shall maintain a record of processed webhook/event identifiers where available. If the same webhook/event is received again, the system shall acknowledge it without repeating the associated business processing.

Webhook receipt and webhook business processing shall be treated as separate steps so that a temporary internal failure does not cause financial duplication when the gateway retries the webhook.

7. Concurrent Request Protection

Duplicate protection shall work under concurrent execution.

If two identical payment requests or webhook events are received simultaneously, database constraints/transactions shall ensure that only one request can create the successful financial posting. The system shall use an appropriate database unique constraint and transactional locking/concurrency mechanism rather than relying only on application-level checks.

8. Payment State Handling

The system shall maintain the following payment states:

INITIATED → PENDING → SUCCESS / FAILED → RETRYING → SUCCESS / FAILED

RETRYING shall indicate that the payment has failed an attempt but remains eligible for automatic recovery under the configured retry/grace-period policy. Each retry attempt shall be recorded separately, while the payment obligation remains associated with the same scheduled instalment.

A payment shall transition from RETRYING to SUCCESS when a retry succeeds. It shall transition to FAILED when the retry limit is exhausted, the grace period expires, or the gateway identifies the payment as non-retrievable.

A payment in RETRYING shall not be treated as a successfully received payment and shall not trigger balance updates, interest-related successful-payment processing, commission eligibility, or other success-based financial processing until it reaches SUCCESS.

A duplicate request referring to a payment in PENDING or RETRYING state shall return the existing payment status and shall not create another payment transaction or retry cycle.

Attempt vs. Payment Status: Each gateway payment attempt shall have its own attempt result (SUCCESS/FAILED), while the overall scheduled payment shall remain in RETRYING while further automatic attempts are permitted. The overall payment shall become SUCCESS only upon a successful attempt and FAILED only when no further automatic recovery is permitted.

9. Amount and Context Validation

Where a gateway callback is received, the system shall validate that the gateway transaction belongs to the expected:

- user;
- plan;
- payment order/reference;
- amount;
- currency;
- payment purpose.

A gateway callback that does not match the original payment context shall be marked for exception/review and shall not automatically credit the user's account.

10. Failed Payment and Retry Interaction

A failed payment attempt shall not prevent a later legitimate retry from succeeding.

However, each retry shall remain linked to the same scheduled payment obligation while each actual gateway transaction shall retain its own gateway reference.

Only an actually successful and unique payment shall be credited.

This shall work consistently with the BRD's configured retry count, retry interval and grace-period rules.

11. Audit Trail

For every duplicate-detection event, the system shall record:

- User ID;
- Plan ID, where applicable;
- Internal Payment Transaction ID;
- Idempotency Key;
- Gateway Order ID;
- Gateway Payment ID;
- webhook/event identifier, where available;
- received timestamp;
- original transaction timestamp;
- duplicate detection reason;
- processing outcome.

Duplicate events shall remain auditable but shall not create additional financial ledger entries.

12. Receipt and Notification Rule

A successful payment receipt and Payment Success notification shall be generated only for the first successfully processed financial transaction.

A duplicate callback/retry shall not generate another receipt, commission, balance update, or Payment Success notification.

13. Database Constraints

At minimum, the database shall enforce uniqueness for:

- Idempotency Key;
- Gateway Payment ID;
- applicable gateway Order ID + successful settlement context;
- processed webhook/event ID, where provided.

The exact uniqueness constraints shall be designed so that legitimate separate instalments/payments for the same Plan are not incorrectly classified as duplicates.

Required acceptance criteria

1. Retrying the same API request with the same Idempotency Key produces one payment transaction only.
2. Receiving the same successful Razorpay webhook multiple times produces one financial posting only.
3. Two concurrent requests for the same payment cannot both credit the user's account.
4. A duplicate cannot generate duplicate referral commission.
5. A duplicate cannot increment the successful-payment count used for commission eligibility more than once.
6. A legitimate subsequent instalment for the same Plan is accepted as a new payment.
7. Every duplicate event is auditable.
8. No duplicate payment can cause more than one unified-ledger financial impact.
9. A Razorpay webhook with an invalid or missing signature is rejected and produces no financial or balance impact.

XXXVII. Franchisee Plan and College Redemption:

The system shall support a Franchisee redemption option under the existing User flow. Admin shall be able to create and manage Franchisee Plans, with each Franchisee Plan containing a **Plan Name** and a **One-Time Deductible Price**.

The User shall be able to view the available Franchisee Plans and select a Franchisee Plan. Upon selection, the system shall display the list of Colleges mapped to that Franchisee Plan. The User shall select a College and submit a Franchisee redemption enquiry to Admin for offline processing.

The system shall compare the **One-Time Deductible Price** with the User's **Available Margin**.

- If **One-Time Deductible Price $\leq$ Available Margin**, the User may submit the Franchisee redemption enquiry normally for Admin review.
- If **One-Time Deductible Price $>$ Available Margin**, the system shall calculate and display the **Shortfall Amount**.

Shortfall Amount = One-Time Deductible Price – Available Margin

- The User shall still be allowed to submit the enquiry with the shortfall, and the enquiry shall move to **AWAITING_SHORTFALL_RESOLUTION**.
- While the shortfall remains unresolved, the complete Franchisee redemption enquiry shall remain financially unprocessed. No portion shall be processed separately and no redemption ledger debit shall be created.
- The shortfall shall be resolved through the applicable offline process, and the required payment details/evidence shall be submitted for Admin verification.
- The shortfall shall be considered resolved only after successful Admin verification.
- After successful verification and approval, the Franchisee redemption shall proceed to financial processing in accordance with the applicable redemption and Unified Financial Ledger rules.
- Upon approval, the applicable deductible amount shall be reflected in the user's redemption records and the **Actual Redeemable Balance** shall be updated in accordance with the approved redemption.
- The selected Franchisee Plan and College shall remain linked to the submitted enquiry and its related financial and audit records.
- Franchisee redemption shall not be subject to the investment Plan Status values or any maturity period. Each approved Franchisee redemption shall be treated as a one-time redemption transaction based on the selected Franchisee Plan and College.

For a Franchisee redemption with a shortfall, the system-side financial settlement shall be limited to the amount available through the user's Redeemable Balance/Available Margin, while the resolved offline shortfall shall be recorded as verification information/evidence. The offline shortfall shall not create a redemption ledger debit unless a separate, explicitly defined financial transaction applies.

XXXVIII. **Upcoming Instalment Reminder:** The system shall send an Email and Push Notification to the User before the scheduled upcoming instalment payment of an ACTIVE Plan. The reminder shall include the applicable Plan, scheduled payment amount, and scheduled payment date. The reminder timing shall be configurable by Admin.

Points to remember:

1. Interest Calculation Method: Admin can create multiple interests out of simple/compound interest or interest calculated by some configurable formula. These interests will be created once and applied to other places wherever required.
2. Interest on Plan: Applicable only at plan maturity.
3. Handling Missed Instalments: A missed instalment shall reduce only the **remaining unpaid principal amount** of the Plan by the amount of the missed instalment. The system shall not reduce previously paid principal amounts. Any interest already accrued shall not be treated as principal for the purpose of this adjustment. Future interest calculations shall use the resulting remaining unpaid principal as the applicable principal balance.

Original Principal	₹10,000
Paid Principal	₹2,000
Remaining Unpaid Principal	₹8,000
Missed Instalment	₹1,000

Adjusted Unpaid Principal ₹7,000

Missed installment → reduce remaining unpaid principal, not the original principal and not previously paid principal.

Boundary Handling: The remaining unpaid principal shall never become negative. When a missed instalment is equal to or greater than the current remaining unpaid principal, the adjusted remaining unpaid principal shall be set to ₹0.00. Any excess amount beyond the remaining unpaid principal shall not create a negative principal balance and shall not be treated as additional missed principal. For cumulative missed instalments, the same rule shall apply after each missed instalment, and once the remaining unpaid principal reaches ₹0.00, no further reduction shall be applied.

Edge Case Example:

Remaining Unpaid Principal ₹800

Missed Instalment ₹1,000

Adjusted Unpaid Principal ₹0

Excess ₹200 is not added as negative principal.

Interest Impact: Future interest calculations shall use ₹0.00 as the applicable remaining unpaid principal once the remaining unpaid principal reaches ₹0.00. No negative interest-bearing principal shall be created.

4. **Educational Benefit Calculation:** Redeemable amount is calculated using one of the Interest Calculation as created by the admin.

5. **Referral Commission Basis:** Commission shall be calculated according to the commission type configured by Admin. For Recurring Commission, commission shall be generated after every four consecutive successful payments. For One-Time Commission, commission shall be generated once on the first successful payment under the applicable Plan.

6. **Payment Gateway Tokenization:**

- Only **Razorpay** shall be used as the payment gateway. Razorpay shall support **secure recurring-payment authorization** for automatic deductions through **tokenized card payments, UPI AutoPay, and eMandate**, as applicable.
- The system shall **not store raw card or other sensitive payment credentials**. Instead, tokenized payment credentials and/or authorized UPI AutoPay/eMandate mandates shall be used for subsequent recurring deductions.
- The backend shall be capable of initiating **authorized automatic deductions through Razorpay's recurring-payment APIs**, subject to applicable payment-network, RBI, NPCI, and mandate-specific limits and conditions.

7. **AutoPay/eMandate Limit Handling:**

- If an automatic deduction cannot be executed because of a payment-network, RBI, NPCI, mandate-specific, bank or gateway limit, the system shall treat the automatic payment attempt as unsuccessful and shall not mark the scheduled payment as successfully received.
- The payment shall then follow the configured **Failed Payment and Retry Management** rules, including the applicable retry schedule and grace period.
- The user shall be notified that the automatic deduction could not be completed and shall be provided the applicable manual payment option where automatic recovery is no longer possible or permitted.
- No interest, commission, ledger credit, balance update, or successful-payment count shall be generated from an unsuccessful AutoPay/eMandate attempt.

****User => Anyone including students who is registered with the platform****

****Referrer => Any registered user****

Redeem Options: (Any payment made to the user will be offline which admin will update manually)

To Redeem a student can opt for courses, refund, re-invest, donate now or "Gadgets and Accessories".

1. Course: If the course fee exceeds the user's Available Margin, the system shall calculate the Shortfall Amount and place the redemption request into **AWAITING_SHORTFALL_RESOLUTION**.

Shortfall Amount = Course Fee – Available Margin

The complete course redemption request shall remain financially unprocessed until the shortfall is resolved offline and the required evidence/details are submitted by the user and verified by an authorized Admin.

The course redemption shortfall shall follow the **same Offline Shortfall Verification procedure defined under Rule XXXIV – Money Mechanics**. The Admin shall record the applicable payment reference/receipt number, payment date, amount, payment method and verification comments, and supporting proof may be uploaded where required.

The shortfall shall not be considered resolved until it has been verified by an authorized Admin in accordance with Rule XXXIV.

The course redemption request shall remain subject to the system-wide **Redemption Request Expiry** rules. A request in **PENDING** or **AWAITING_SHORTFALL_RESOLUTION** shall expire after the configured maximum period if the required action has not been completed. Upon expiry, the request shall move to **EXPIRED** and any associated financial and inventory reservation shall be released in accordance with the applicable expiry rules.

The expiry period shall be the same Admin-configured period applicable to other redemption requests and shall not use a separate course-specific timeout.

After successful Admin verification and approval, the system shall process the course redemption in accordance with the applicable redemption and ledger rules.

If the shortfall is not resolved before the applicable expiry period, the course redemption request shall expire/cancel and no redemption ledger debit shall be created.

2. **Reinvestment**: Reinvesting the amount redeemable balance will be updated accordingly. Before maturity only redeemable balance should be available.
 3. **Donate**: Donation of an either complete or partial amount, an enquiry form will be presented to set amount and fill other relevant details.
 4. **Refund**: An enquiry will be generated for either partial or complete refund which will be addressed by the admin.
 5. **Accessories and Gadgets**: User can use a partial or complete amount of their redeemable balance to raise an enquiry for available gadgets and accessories. A view shall display the available items and their respective prices. No tax/GST or shipping cost required for this feature. The list should be simply displaying the price for each gadget.
- When the user selects one or more items, the system shall calculate the **Total Requested Redemption Amount** and compare it with the user's **Available Margin**.
- If **Total Requested Redemption Amount $\leq$ Available Margin**, the user may submit the redemption request normally.
 - If **Total Requested Redemption Amount $>$ Available Margin**, the system shall display:
 - Actual Redeemable Balance
 - Available Margin
 - Total Requested Redemption Amount
 - Shortfall Amount

Shortfall Amount = Total Requested Redemption Amount – Available Margin

An amount exceeding the Available Margin shall **not prevent the user from submitting the request**.

The user may submit the request with the shortfall.

The request shall then move to **AWAITING_SHORTFALL_RESOLUTION**.

While the shortfall remains unresolved:

- the full redemption request shall remain financially unprocessed;
- no portion of the request shall be processed separately;
- no redemption ledger debit shall be created;
- the Actual Redeemable Balance shall not be reduced due to the redemption;
- the user shall be required to resolve the shortfall offline;
- Admin shall verify the submitted shortfall resolution before approval.

Only after successful Admin verification and approval shall the redemption proceed to financial processing in accordance with the applicable redemption rules.

Example:

Actual Redeemable Balance = ₹10,000

Available Margin = ₹10,000

Selected Gadgets = ₹12,000

Shortfall = ₹2,000

The system shall allow submission, display the ₹2,000 shortfall, and move the request to **AWAITING_SHORTFALL_RESOLUTION**. It shall not debit ₹10,000 separately.

Any pending gadget request that is subsequently modified shall be replaced by the revised total requested amount. The previous pending reservation shall not be counted again when recalculating Available Margin or Shortfall.

6. Franchisee: Admin shall be able to create and manage Franchisee Plans. Each Franchisee Plan shall contain a Plan Name and a One-Time Deductible Price. The User shall be able to view the available Franchisee Plans and select a Franchisee Plan.

After selecting a Franchisee Plan, the system shall display the Colleges mapped to that Franchisee Plan. The User shall select a College and submit a Franchisee redemption enquiry to Admin for offline processing.

The system shall compare the One-Time Deductible Price with the User's Available Margin.

- If the One-Time Deductible Price $\leq$ Available Margin, the User may submit the enquiry normally for Admin review.
- If the One-Time Deductible Price $>$ Available Margin, the system shall calculate and display the Shortfall Amount.

Shortfall Amount = One-Time Deductible Price – Available Margin

- The User shall still be allowed to submit the enquiry with the shortfall. The enquiry shall move to **AWAITING_SHORTFALL_RESOLUTION**.

- While the shortfall remains unresolved, no portion of the Franchisee redemption shall be financially processed and no redemption ledger debit shall be created.

- The shortfall shall be resolved offline and the required payment details/evidence shall be submitted for Admin verification.

- Only after Admin verification and approval shall the Franchisee redemption proceed to financial processing in accordance with the applicable redemption and Unified Financial Ledger rules.

After every redemption balance available to redeem will be shown as margin which is the available balance left to request something. Whereas actually redeemable balance wouldn't be affected by the raised enquires.

For example, if ₹2000 as redeemable balance and opt for gadgets with price 200, then the available margin will be 1800 whereas the redeemable balance will remain same unless admin approves the enquiry. Once approved the redeemable balance gets reduced by 200 and updated accordingly.

As for enquiry admin can raise concerns if there are any and share then in form of messages against the raised enquiry.

The Admin shall not directly edit the user's Reward Points or Actual Redeemable Balance. Upon approval of an eligible redemption, the system shall automatically calculate and apply the applicable financial deduction through the Unified Financial Ledger and redeemable balance would get updated accordingly.

These balances won't have expiry.

Note: Any enquiry generated by the user when approved will result in deduction of the redeemable balance. Inventory should update once request for accessories and gadgets raised in enquiry gets approved by the admin.

Admin Area:

Admin user should be able to view and manage calculations plans, users, enquiries an accessories.

Admin will be able to access and generate all kinds of reports related to payment, referrals, enquiries, payments made to user, accessories and gadget inventory, etc.

User Management:

- Registration process
- Login
- OTP verification using email
- Email verification
- Forgot Password
- Change Password
- Social Login: Users shall be able to register and log in using administrator-configured social identity providers (e.g., Google and/or other approved OAuth 2.0/OpenID Connect providers). The system shall use the provider's authorization flow and shall not store the user's social-provider password. On successful authentication, the system shall create or link the user account using the verified provider identity/email. A user shall not be assigned duplicate accounts when the same verified email is already registered. Social-login users may set a local password later through the applicable account/password flow.

- **Mobile Number:** The user's mobile number shall be stored as part of the user's profile/contact information and shall be available to Admin in User Management. The mobile number shall be associated with the user's account and shall be editable by the user subject to system validation. The mobile number shall not be used for login, 2FA, password reset, or other authentication OTP purposes. A mobile number shall be marked as verified only after successful completion of the configured verification process.

Roles:

- **Admin:** Can manage the webapp features. The admin should be able to handle payout time which will be only an information shared with user in the form of notification as well as in their dashboard.

Responsibilities:

- Dashboard
- User Management
- Plan Management
- Interest/Reward configuration
- Payment monitoring
- Referral commission and code configuration
- Reward point configuration
- University management
- Course management
- Gadget & Accessories management
- Redemption request management
- Enquiry management
- Notifications
- Reports
- Audit logs
- Site settings
- **User:** Can only create a profile and perform user related activities as per the documentation.

Responsibilities:

- Register/Login
- View available plans
- Subscribe to one or multiple plans
- Make payments
- Manage auto-debit mandate
- View investment summary
- View redeemable balance
- Redeem for (all of them will result in an enquiry to the admin)
 - Courses
 - Refund
 - Reinvestment
 - Donation
 - Franchisee
 - Gadgets & Accessories
- View reward points
- Generate and share referral code
- View referred users
- View referral earnings
- Download payment receipts
- Receive notifications
- Update profile and payment details

Donation Financial Settlement: A Donation request shall remain an enquiry/request until Admin approval. Before approval, the Actual Redeemable Balance shall remain unchanged and no redemption ledger debit shall be created. After approval, the approved donation amount shall be posted as a redemption transaction in the Unified Financial Ledger and deducted from the user's Redeemable Balance. For a partial donation, only the approved donation amount shall be deducted.

Franchisee Plan Management (Admin):

- Franchisee Plan Name
- One-Time Deductible Price

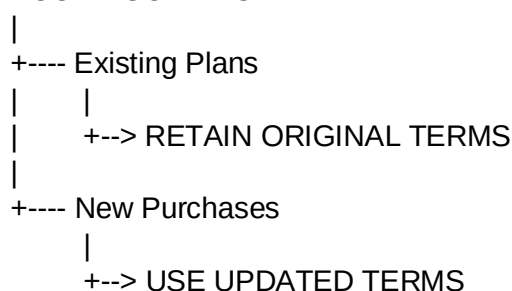
- College Mapping to Franchisee Plan

Plan Management (Admin):

- Investment Tenure
- Monthly/Lumpsum
- Interest
- Reward - a percentage configured by the Admin that shall be applied to the applicable course fee for an approved course redemption to calculate the reward points/amount. $\text{Reward Amount} = \text{Applicable Course Fee} \times \text{Configured Reward Percentage}$
- Plan active/inactive
- List of the amount to be invested
- Payment Frequency
- **Plan Configuration Update:** Any administrative modification to an existing plan's configurable terms, including interest rate, tenure, reward percentage, commission percentage, or other applicable plan parameters.
- **Existing Plan Terms:** When an Admin updates the configuration of a plan, the updated configuration shall apply only to new plan purchases created after the effective configuration change. Existing plans shall continue to use the terms captured at the time of their purchase unless an explicitly defined migration operation is performed.
- **Reward Percentage Scope:**
The Admin shall be able to configure the Reward Percentage either **for each plan separately** or **as a common Reward Percentage applicable across all plans**.
The Reward Percentage applicable to a course redemption shall be determined from the configuration applicable to the user's plan at the time of the relevant plan purchase/redemption, consistent with the existing Plan Configuration and Existing Plan Terms rules.

• Recommended flow

PLAN CONFIGURATION

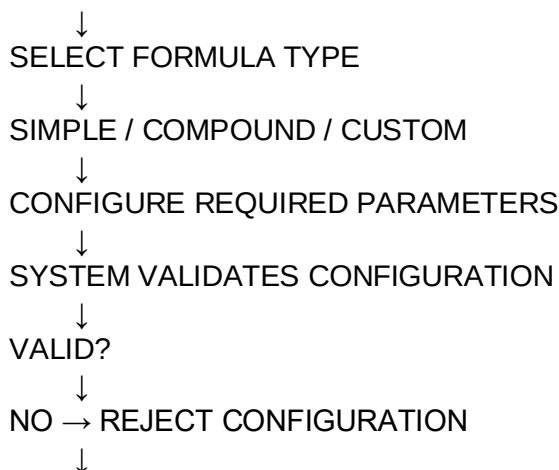


Interest Engine:

The system shall provide an Interest Engine through which the Admin can configure and assign supported Interest Calculation Methods to plans. The detailed formula types, parameters, validation, versioning, snapshot/locking behaviour, and ledger requirements shall be governed by **Rule XXVI – Interest Calculation Formula Configuration**.

Simplified Interest Configuration Flow:

ADMIN CONFIGURES INTEREST METHOD



YES
↓
SELECT PLANS
↓
APPLY INTEREST METHOD
↓
SNAPSHOT FORMULA + PARAMETERS
↓
LOCK PLAN-SPECIFIC CONFIGURATION
↓
CALCULATE INTEREST
↓
RECORD CALCULATION IN LEDGER
↓
STORE METHOD / VERSION REFERENCE

Important clarification:

Plan A
Interest Rate = 10%
Method Version = V1
↓
Interest accrued using V1
↓
Admin changes configuration to 12%
↓
New Method Version = V2
↓
Plan A continues using V1
↓
V2 applies only when assigned to eligible/new plans

Payment Module:

- Payment Gateway
- Auto debit via credit or debit card, UPI, Net banking, digital wallet. Payment method can be changed after creation of an ACTIVE or DISCONTINUED Plan, subject to the applicable payment gateway validation and authorization requirements.
- Receipts
- Failed payment retries
- Grace period
- Maximum retries
- Card expiry handling
- Change payment method
- Redeemable Balance - Display the user's redeemable balance as a monetary amount in **INR (₹)**. The Redeemable Balance shall represent the actual monetary value available for eligible redemption. It shall remain distinct from **Reward Points**, which are maintained separately as a course-only benefit. **Redeemable Balance ≠ Reward Points**: Redeemable Balance is a monetary INR value and is used for eligible financial redemptions. Reward Points are separate non-cash points used only for eligible future course-fee adjustment and cannot be withdrawn.
- Payment history
- Duplicate payment handling

Referral Module:

- Referral Relationship Management
- Referral Validity Period Management

- **Referral Code Management:** Each user shall have one active referral code at a time. The user shall be able to regenerate/rotate their referral code when required, such as when the code is compromised or unintentionally shared. Regeneration shall deactivate the previous code for all new referral registrations and generate a new unique code. Existing referral relationships created using the previous code shall remain unchanged and valid. A deactivated referral code shall not create any new referral relationship. Referral codes shall be exactly 8 characters long and use only uppercase letters (A–Z) and digits (0–9). Each active referral code shall be unique across all users. Code generation shall use secure random generation, with database-level uniqueness enforcement.
- **Referral Cancellation Management**
- **Commission reports**

Accessories / Gadget Module

- **Gadget & Accessories Management:** Admin can manage categories, items, prices, stock and inventory.
- **Catalogue:** User can view available gadgets/accessories and their prices. No GST/tax or shipping cost is added.
- **Selection:** User can select one or more items and raise a redemption enquiry using partial or complete redeemable balance.
- **Balance Check:** System calculates Total Requested Redemption Amount and compares it with Available Margin.
- **Normal Request:** Where the requested amount is within Available Margin, the enquiry can be submitted for Admin review.
- **Shortfall Request:** Where the requested amount exceeds Available Margin, the system displays the Shortfall Amount and allows submission in AWAITING_SHORTFALL_RESOLUTION.
- **Offline Resolution:** Any shortfall/payment is resolved offline and the required details/evidence are submitted for Admin verification.
- **Approval:** Only after Admin verification and approval does the enquiry proceed to financial processing.
- **Financial Update:** On approval, the system updates the Unified Financial Ledger and reduces the Actual Redeemable Balance.
- **Inventory:** Inventory is reserved for the enquiry and converted to a stock deduction only after approval; reservations are released on cancellation, rejection or expiry.
- **Admin Communication:** Admin can raise concerns and communicate messages against the enquiry.
- **Modification:** A modified pending enquiry replaces the previous requested amount/reservation and recalculates Available Margin and Shortfall.
- **Expiry:** Pending enquiries remain subject to the system-wide redemption expiry rules.
- **Franchisee Module:**
 - Franchisee Plan Management
 - One-Time Deductible Price Management
 - College Mapping
 - Franchisee Plan Selection
 - College Selection
 - Franchisee Redemption Enquiry
 - Available Margin Check
 - Shortfall Calculation
 - Offline Shortfall Resolution
 - Admin Verification and Approval
 - Redemption Financial Processing
 - Enquiry Status and History

Notification System:

Need:

- Email
- Push Notification

Events:

- Registration

- Payment Success
- Payment Failure
- Plan Maturity
- Referral Earned
- Refund Processed
- Admin Messages
- Promotional Notifications
- Upcoming Instalment Reminder: The system shall send an Email and Push Notification to the User before the scheduled upcoming instalment payment of an ACTIVE Plan. The reminder shall include the applicable Plan, scheduled payment amount, and scheduled payment date. The reminder timing shall be configurable by Admin.

Student Dashboard:

- Total invested
- Total interest
- Redeemable balance
- Reward points
- Active plans
- Closed plans
- Next deduction
- Referral earnings
- Transaction graph

Admin Dashboard:

- Total users
- Active plans
- Today's payments
- Pending refunds
- Pending enquiries
- Referral payout
- Revenue
- Failed payments
- Upcoming maturities

Reports:

- Payment Report
- Interest Report
- Referral Report
- Plan Report
- Refund Report
- Reward Report
- Enquiry Report
- Revenue Report
- Audit Report
- User Report
- Export:
 - Excel
 - PDF
 - CSV

Security:

- Password encryption
- JWT/OAuth
- Role Based Access
- Device login
- 2FA
- Login history
- Audit logging

APIs:

- Login
- Register
- Plans

- Payments
- Referral
- Rewards
- Redeem / Franchisee
- Reports
- Dashboard
- Notifications

Database Design:

- Master
 - Users
 - Plans (or Plan Configurations)
 - Interest
 - Inventory
 - Franchisee Plans
 - Colleges
 - Franchisee Plan–College Mapping
- Transactional
 - Investments
 - Financial Transactions / Unified Ledger Redemptions
 - Redemptions
 - Notifications
 - Audit Logs

Commission-specific fields can exist within the unified transaction structure.

Transaction Type = COMMISSION

Referrer User ID

Referred User ID

Commission Amount

Commission Status

Accrual Date

Approval Date

Credit Date

University–Course Relationship:

Each Course shall be associated with **one University**, and one University may have **multiple Courses**.

Relationship: **University 1 → Many Courses**

The Course record shall contain a mandatory University ID reference to its associated University.

A Course cannot be created or maintained without an associated University. When displaying or managing Courses, the system shall identify the associated University.

A University may have zero, one, or multiple Courses.

Non-Functional Requirements:

Performance

- Standard user-facing pages and API operations should load/respond within **2 seconds under normal operating conditions**.
- The PWA should be optimized for fast loading and efficient mobile performance.

Availability

- The production system shall provide **99.9% monthly uptime**.
- Planned maintenance shall be excluded from uptime calculation.

PWA & Responsiveness

- The application shall be responsive across mobile, tablet and desktop devices.
- The PWA shall support installation on supported devices.
- Service-worker caching shall be used where appropriate to improve loading and repeat access.

Security

- All application communication shall use **HTTPS**.
- Authentication, authorization and payment-related data shall be handled securely.

Scalability

- The system architecture should support increasing users and transaction volume without major application redesign.

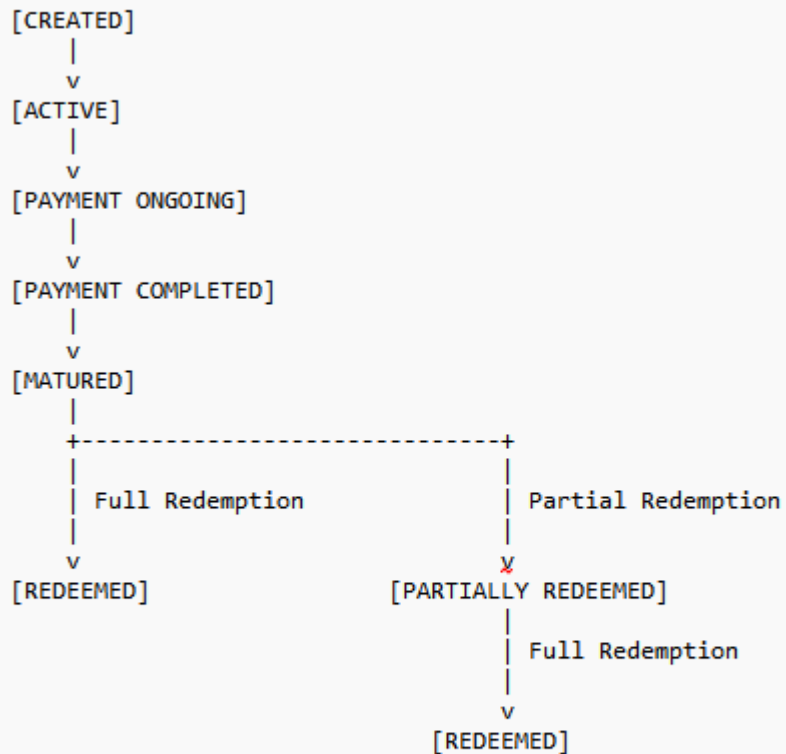
REWARD POINTS: Course-only benefit

REFERRAL COMMISSION: Commission Records / View

Sequence of below diagrams are not related to the sequence of business requirement rules.

1. MASTER PLAN STATE DIAGRAM

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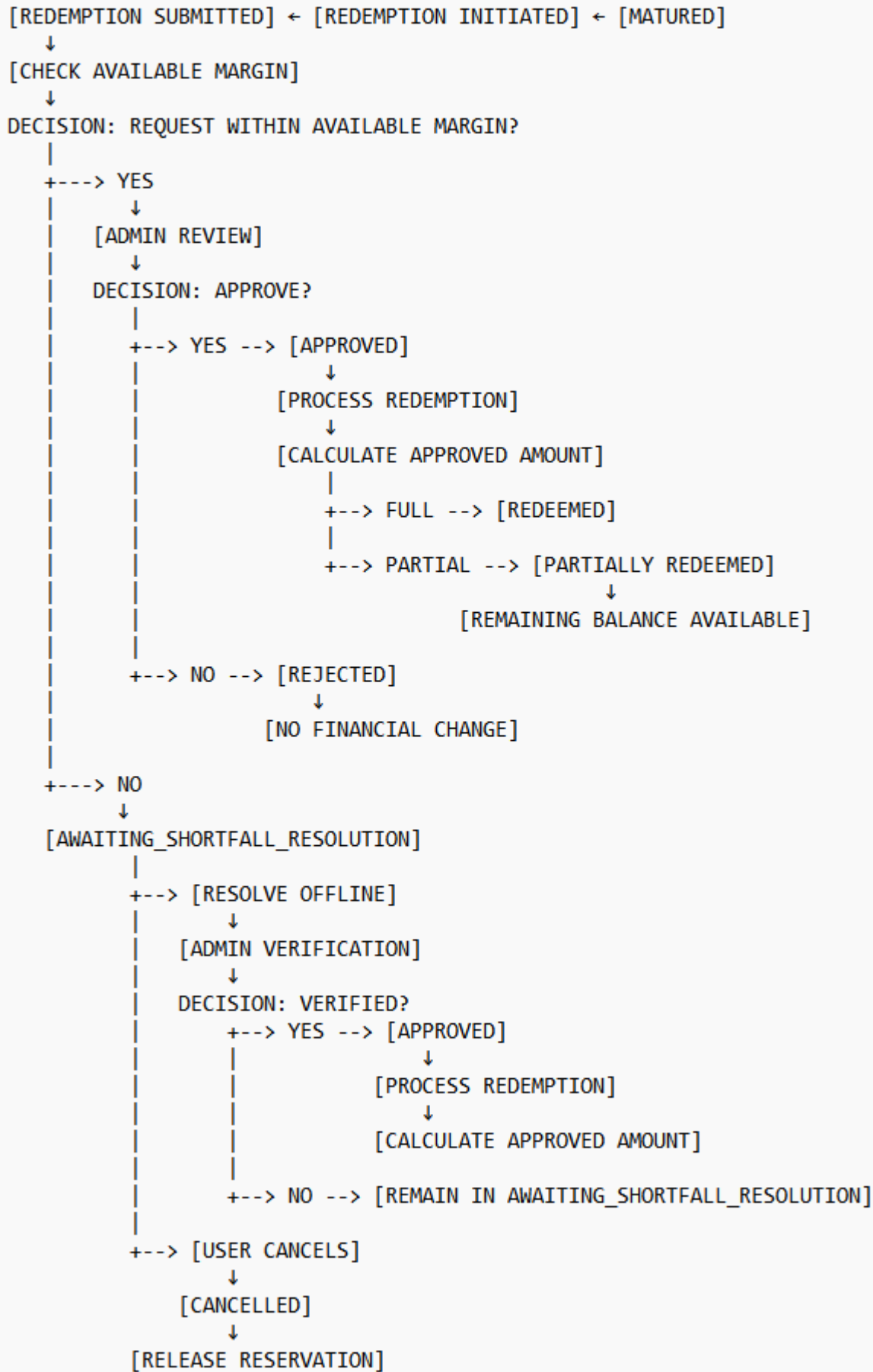


STATE RULES

- PARTIALLY_REDEEMED is a first-class plan state.
- A partially redeemed plan can later become REDEEMED.
- REDEEMED means the remaining redeemable balance is zero.

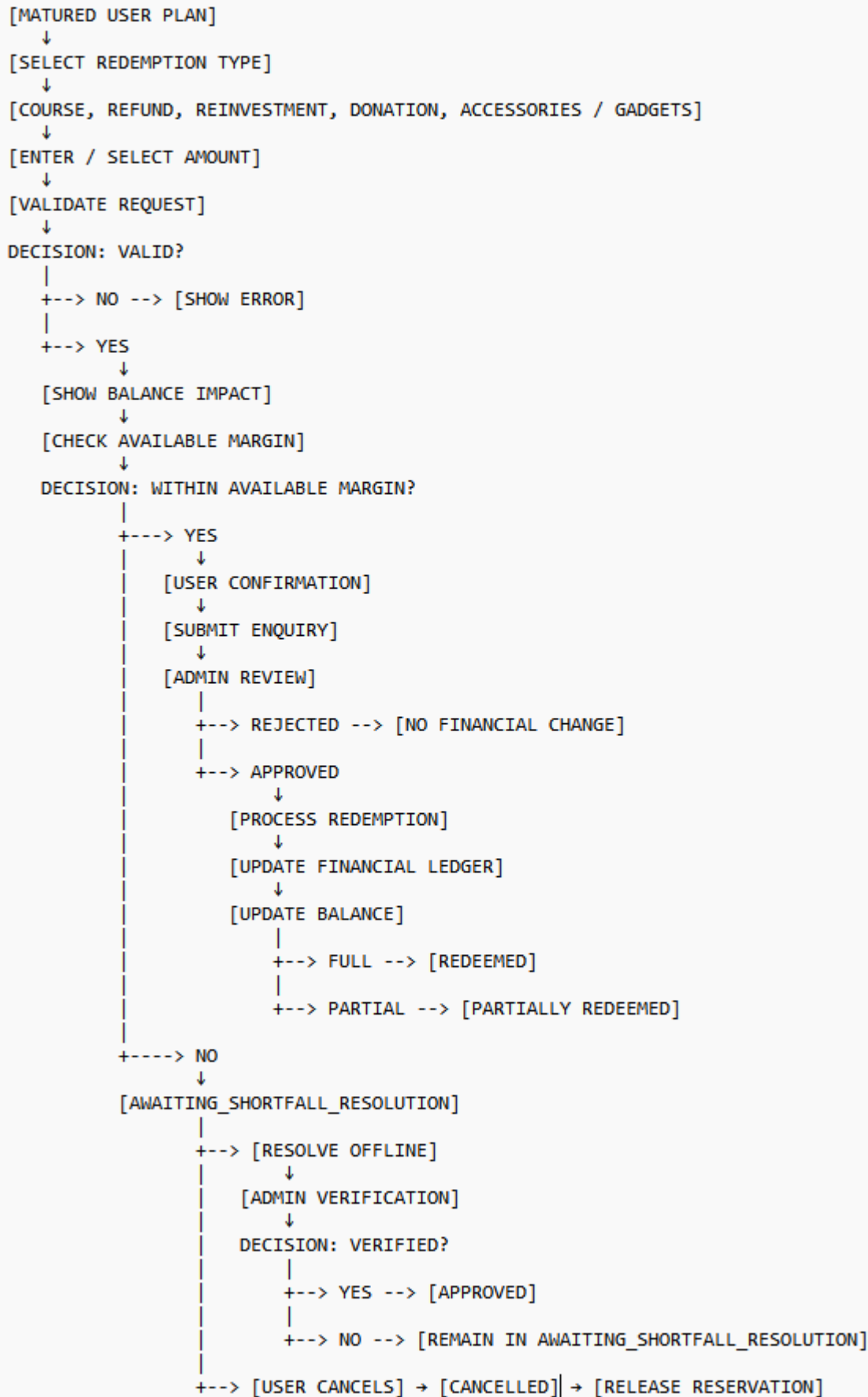
2. REDEMPTION STATE FLOW

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3. WORKFLOW BY REDEMPTION TYPES

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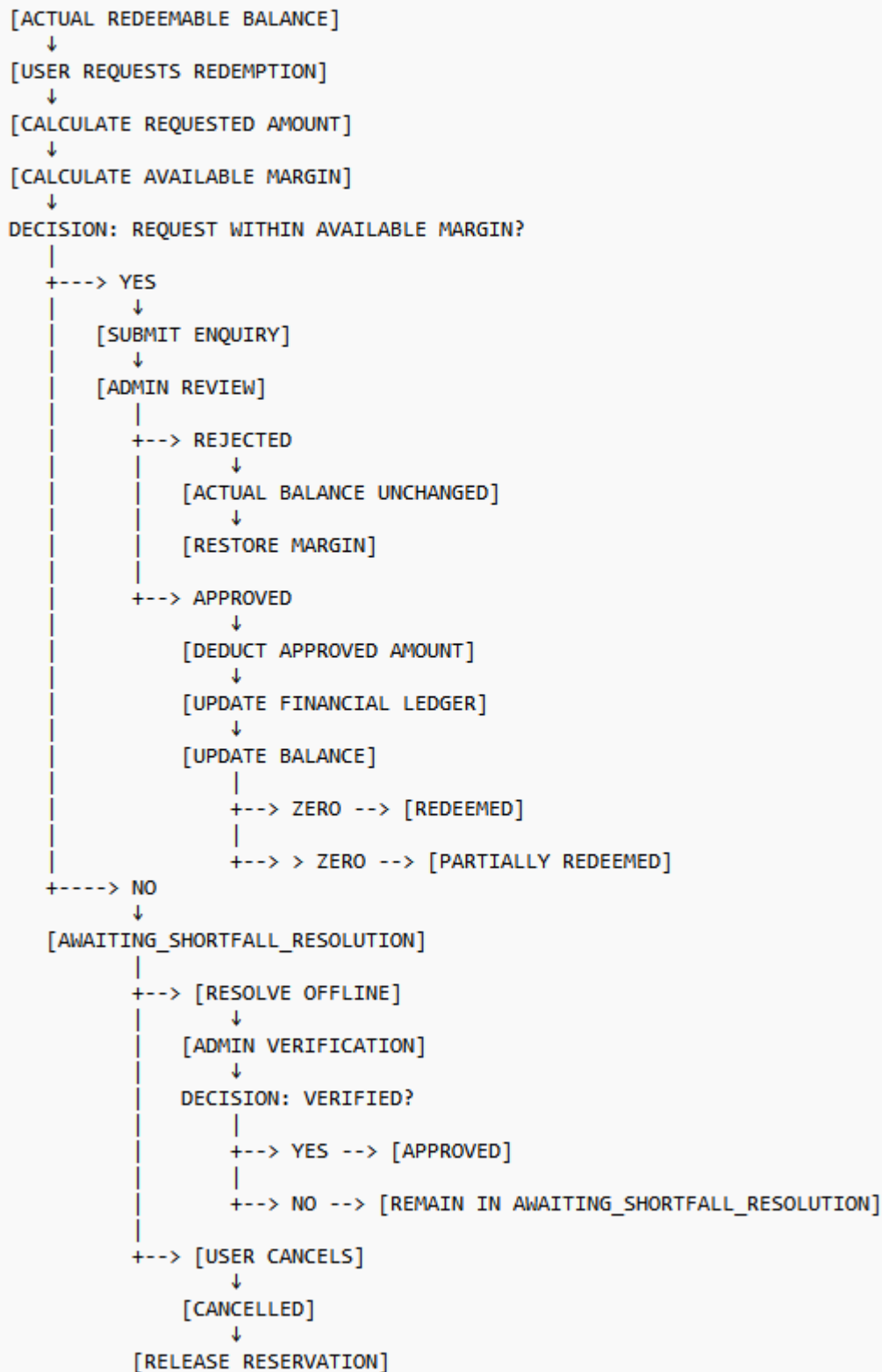


4. ENHANCED REDEMPTION FLOW WITH BALANCE IMPACT



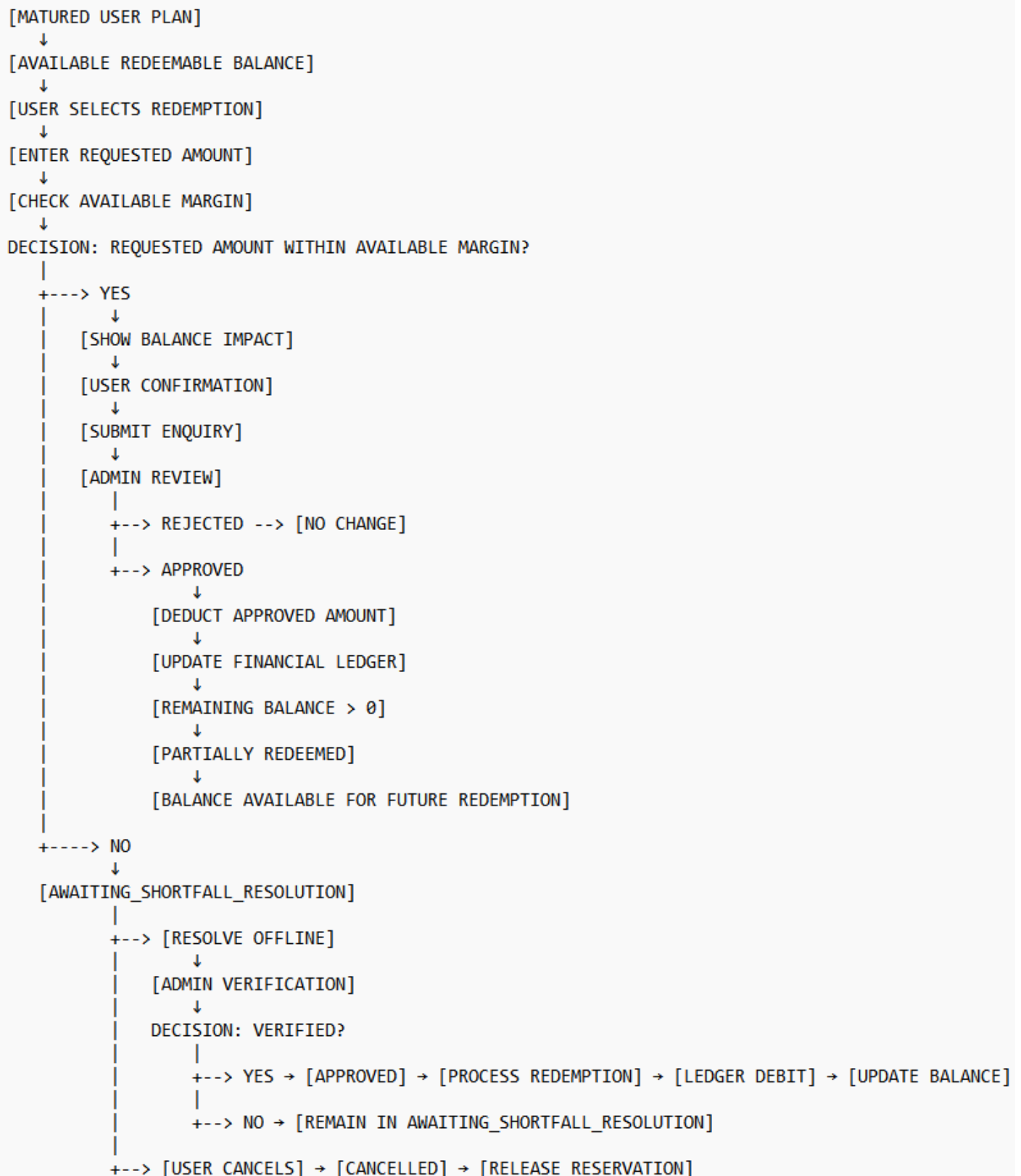
5. BALANCE IMPACT FLOW

=====



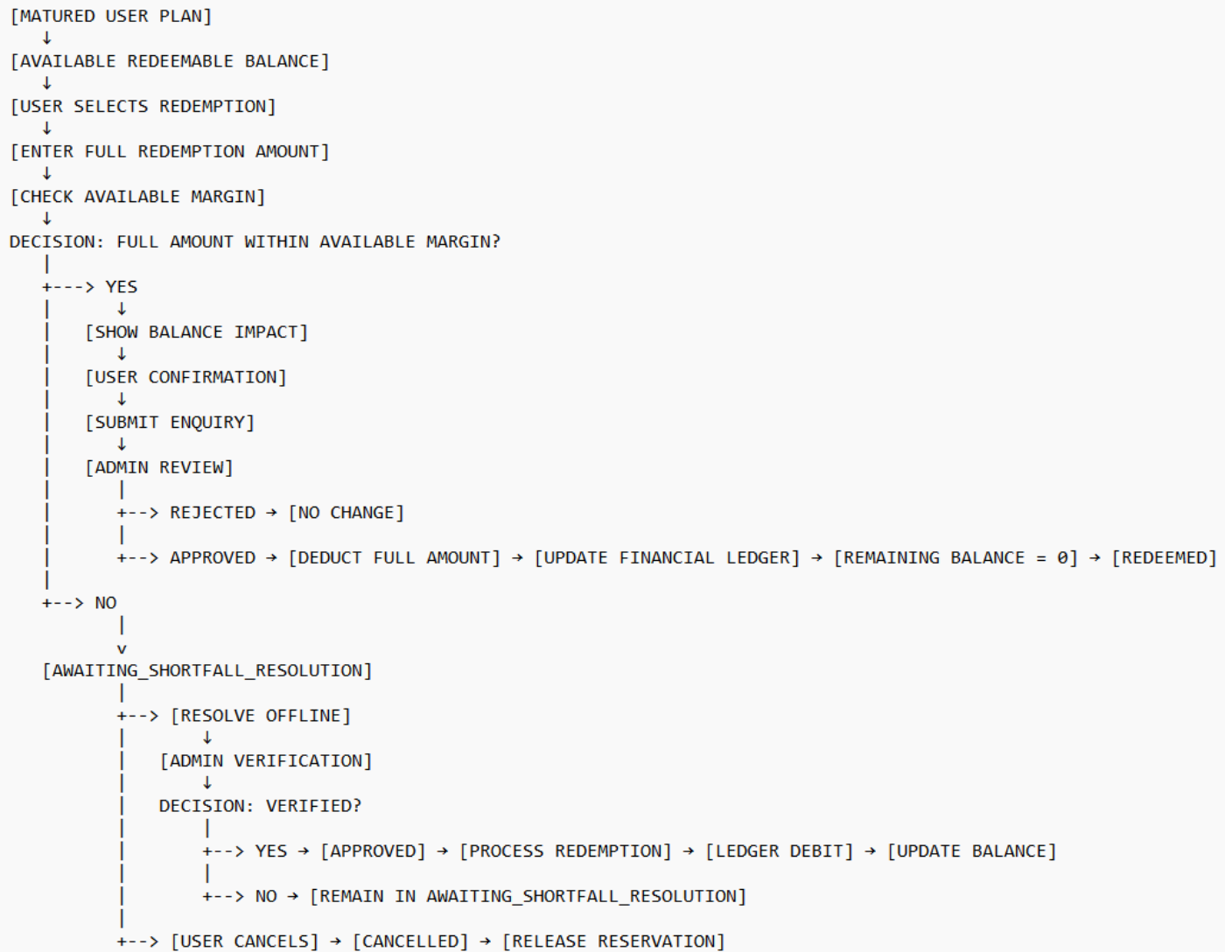
6. PARTIAL REDEMPTION FLOW

=====



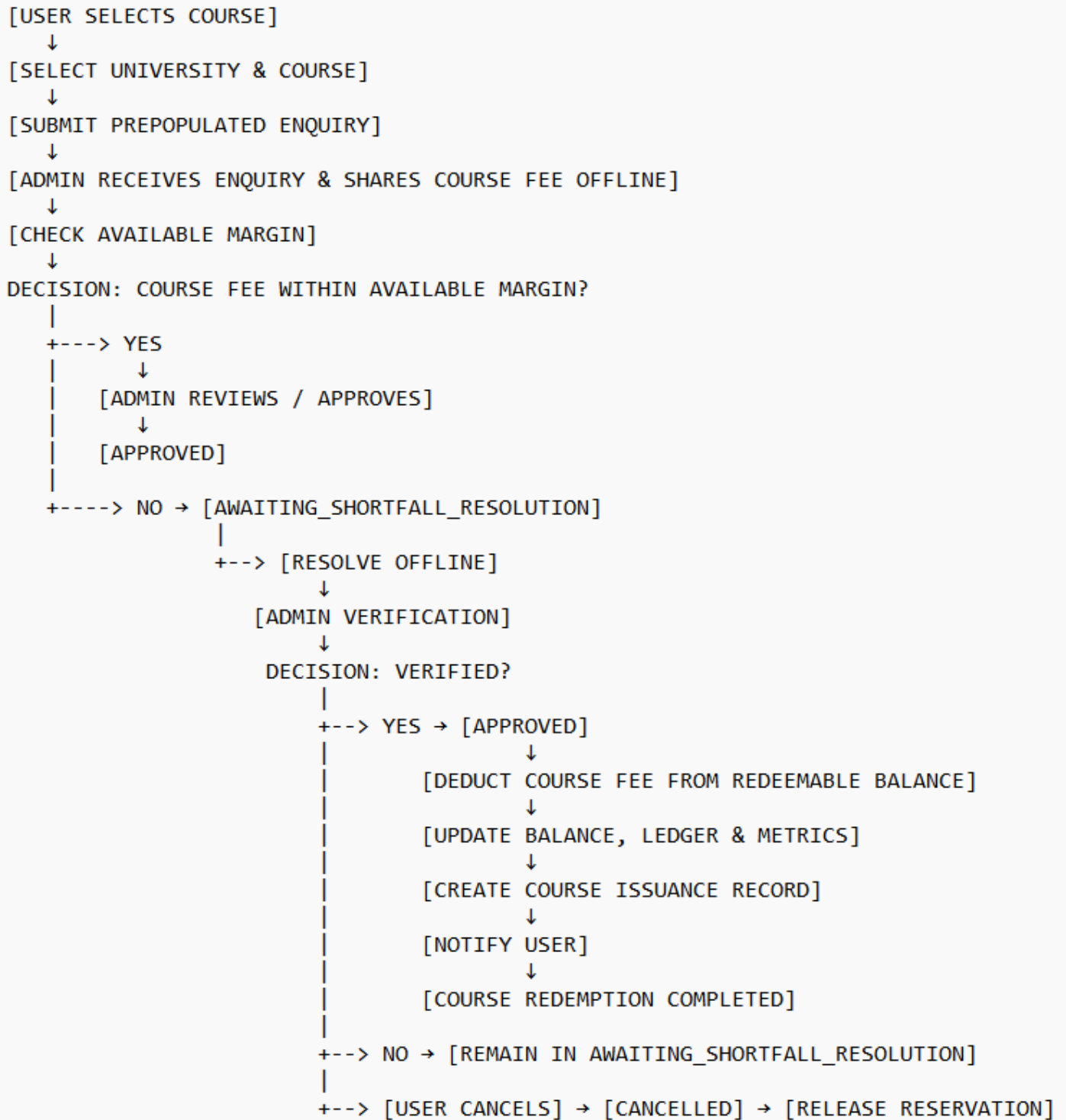
7. FULL REDEMPTION FLOW

=====



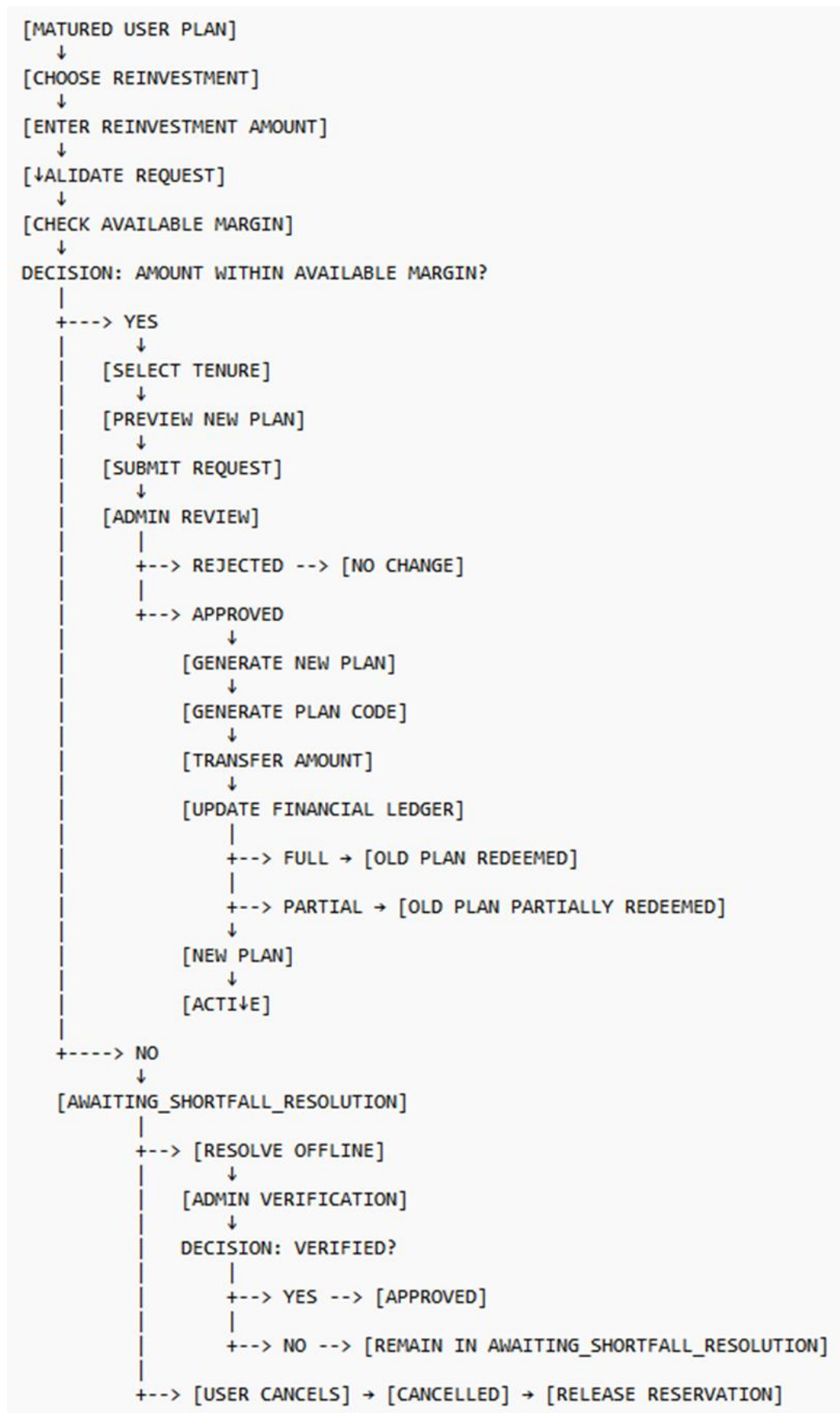
8. COURSE REDEMPTION

=====



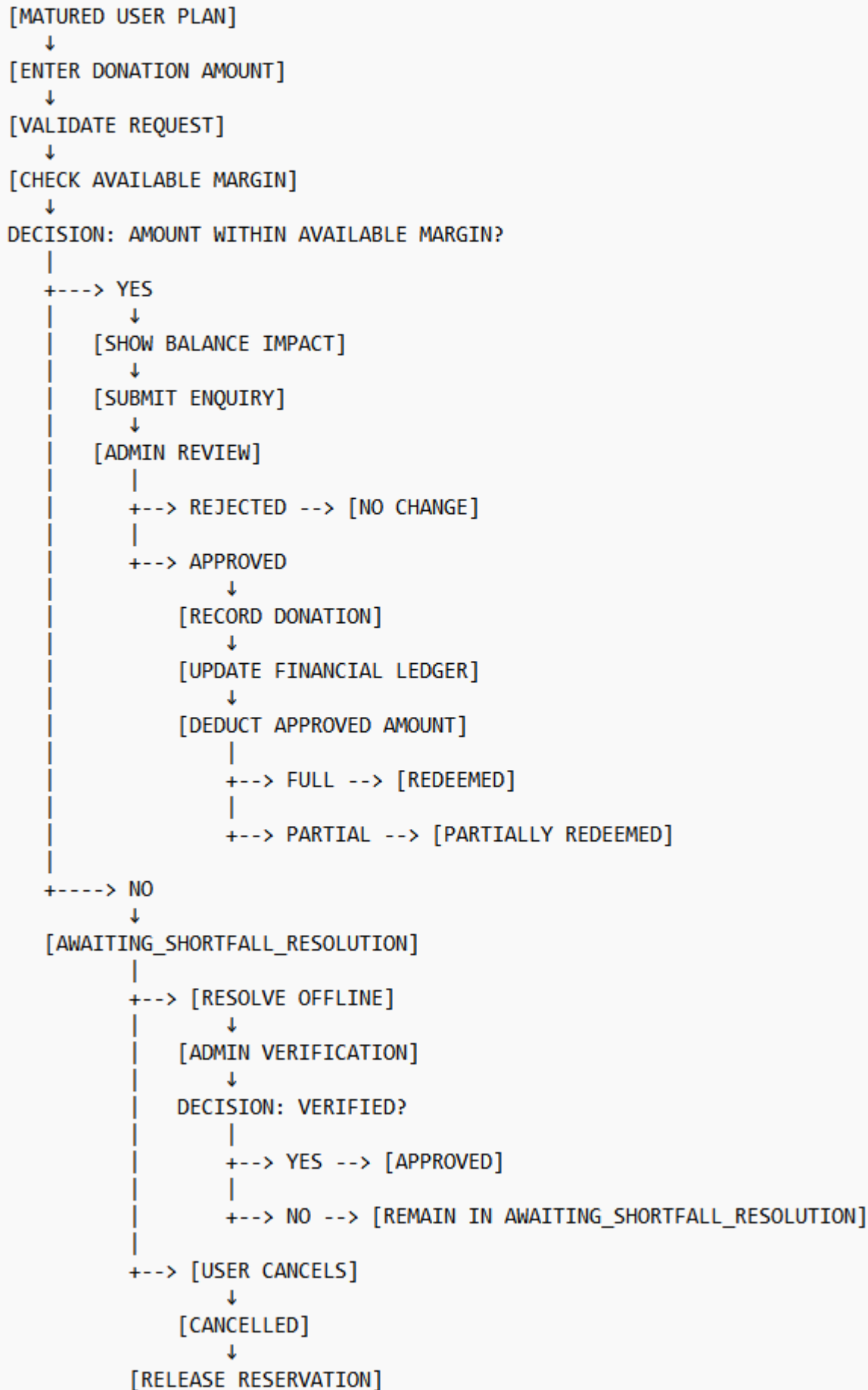
Here course fee comparison is made by admin by referring to the available redeemable balance and then manually adjusting redeemable balance after approving the course issuance.

9. REINVESTMENT FLOW

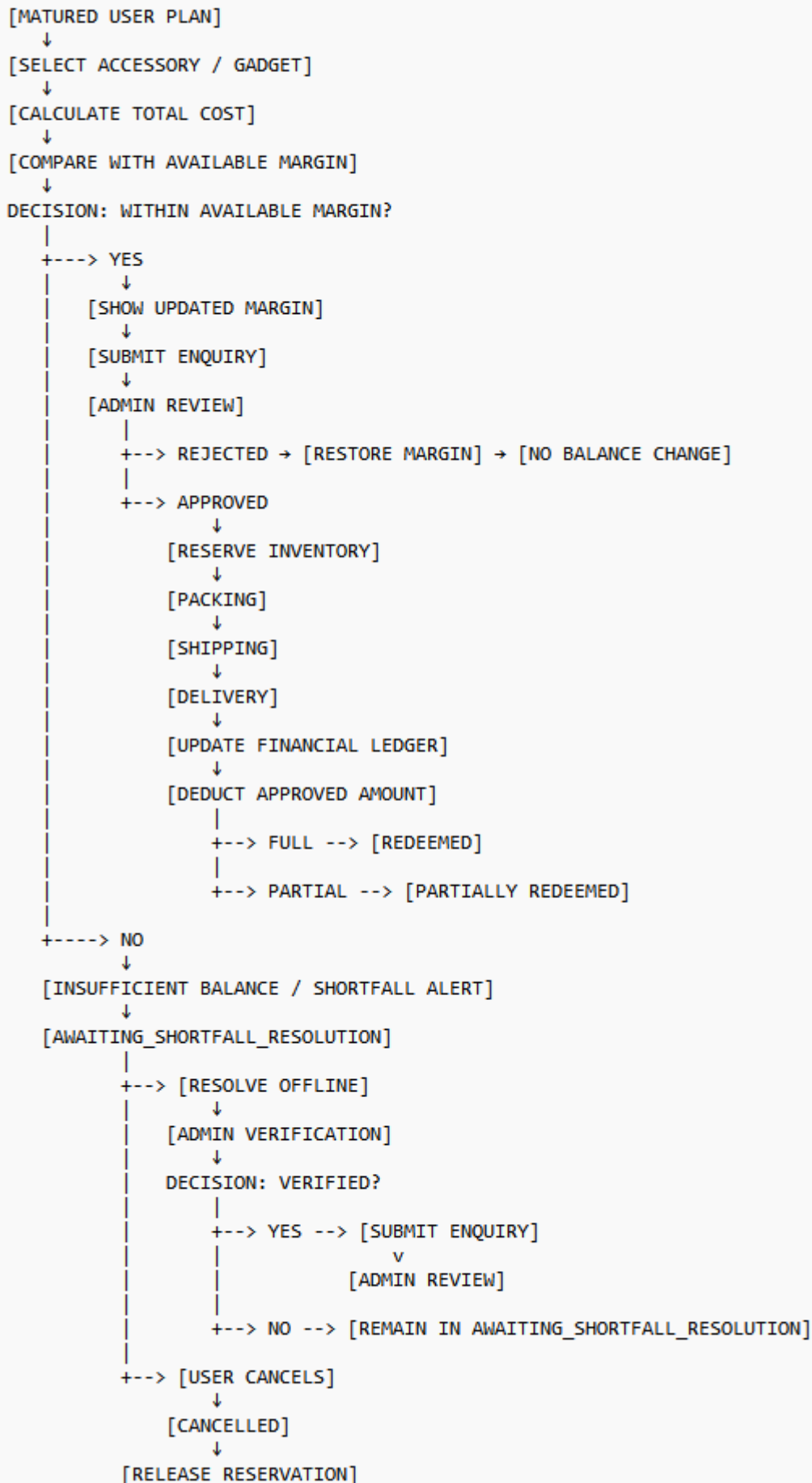


RULE: Reinvestment is optional. Partial reinvestment leaves the remaining balance available.

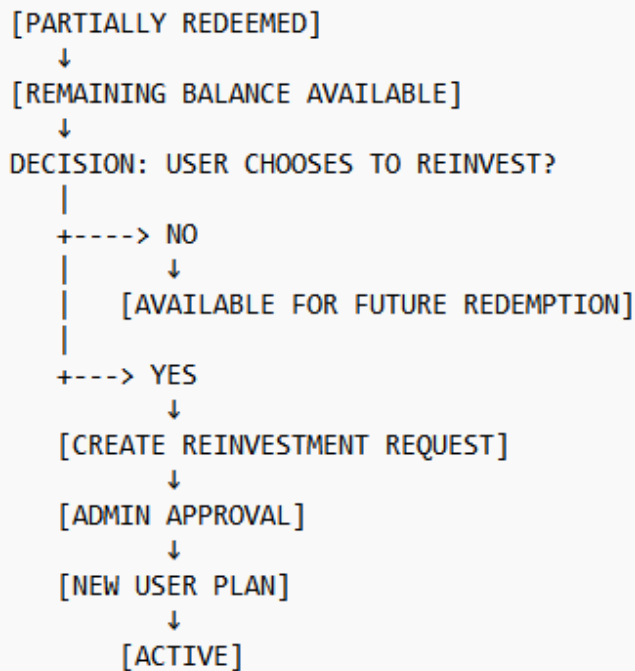
10. DONATION REDEMPTION FLOW



11. ACCESSORIES / GADGETS REDEMPTION FLOW



12. REMAINING BALANCE / OPTIONAL REINVESTMENT



RULE: The remaining balance may be kept for future eligible redemption or optionally reinvested. Reinvestment is not mandatory.

13. Financial Ledger Flow:

Ledger data:

- Ledger Entry ID
- User ID / Plan ID reference
- Transaction Type
- Redemption Type, where applicable
- Requested Amount, where applicable
- Approved Amount, where applicable
- Transaction Amount
- Transaction Status
- Balance Before Transaction
- Balance After Transaction
- Interest Calculation Method ID, where applicable
- Interest Configuration/Parameter Version, where applicable
- Referrer User ID, where applicable
- Commission Reference, where applicable
- Approver User ID, where applicable
- Approval Timestamp (UTC), where applicable
- Transaction Timestamp (UTC)
- Description/Reason, where applicable

Ledger Data Type, Precision and Timezone

- Ledger Entry ID / User ID / Plan ID / Reference IDs: Stored as unique system identifiers.
- Transaction Type / Redemption Type / Transaction Status: Stored as controlled system-defined values.
- Requested Amount / Approved Amount / Transaction Amount / Balance Before Transaction / Balance After Transaction: Stored and calculated in INR with two decimal places (paise).
- Percentage/rate values used in ledger calculations: Stored with up to two decimal places.
- Transaction Timestamp / Approval Timestamp: Stored as date-time values with timezone information. The system shall use UTC for storage and display timestamps to users/admins in the configured application timezone.

- **Commission and interest amounts:** Follow the same INR two-decimal monetary precision and rounding rules defined in Rule XXXIII and Rule XXXIV.

Dashboard Commission Metric Definitions

Dashboard commission metrics shall be derived from the **Unified Financial Ledger** and shall use the Commission Status/Lifecycle defined in this BRD.

1. Referral Earnings – Student/User Dashboard

“Referral Earnings” shall represent the total commission that has been financially credited to the Referrer's commission balance or has already been withdrawn.

Referral Earnings = Sum of Commission Amounts where Commission Status is:

- **CREDITED**
- **AVAILABLE FOR WITHDRAWAL**
- **WITHDRAWN**

Commission in **ACCRUED** or **APPROVED** status shall not be included in Referral Earnings because it has not yet been credited to the Referrer's commission balance.

The same commission transaction shall not be counted more than once across these statuses.

2. Referral Payout – Admin Dashboard

“Referral Payout” shall represent the total commission amount that has actually been withdrawn/paid out to Referrers.

Referral Payout = Sum of Commission Amounts where Commission Status = WITHDRAWN

Commission in ACCRUED, APPROVED, CREDITED, or AVAILABLE FOR WITHDRAWAL status shall not be included in Referral Payout.

3. Source of Calculation

These metrics shall be calculated from the Unified Financial Ledger/Commission Transactions and shall not be maintained as separate manually updated financial balances.

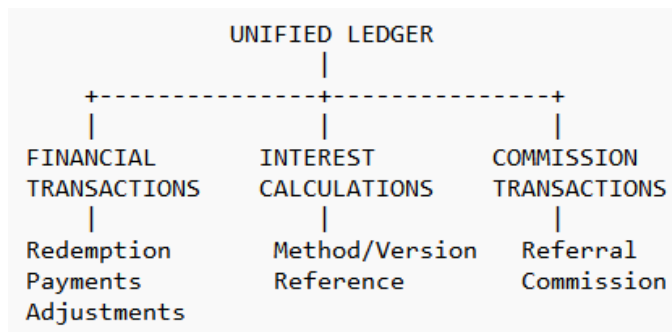
The calculation shall be reproducible from the underlying commission transaction records using the Commission Status, Commission Amount and applicable transaction identifiers.

4. Lifecycle Reference

The commission lifecycle used for these dashboard calculations is:

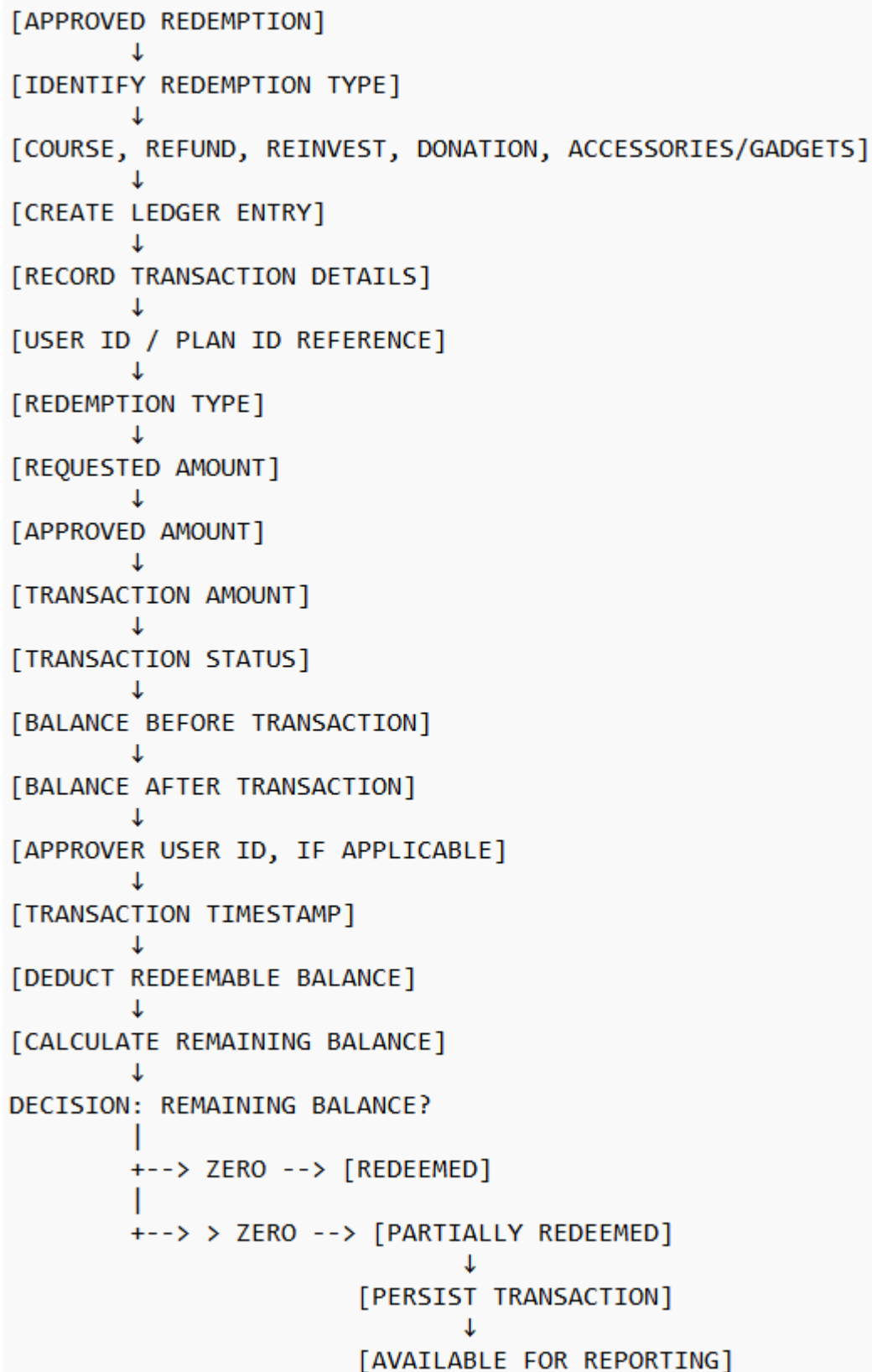
ACCRUED → APPROVED → CREDITED → AVAILABLE FOR WITHDRAWAL → WITHDRAWN

The metric definitions above shall remain aligned with this lifecycle.



LEDGER ENTRY

```
|
+-- Common fields
|   Ledger Entry ID
|   User ID
|   Plan ID
|   Transaction Type
|   Transaction Amount
|   Status
|   Balance Before
|   Balance After
|   Transaction Timestamp
|
+-- Redemption-specific
|   Redemption Type
|   Requested Amount
|   Approved Amount
|
+-- Interest-specific
|   Interest Calculation Method ID
|   Configuration/Parameter Version
|
+-- Commission-specific
|   Referrer User ID
|   Commission Reference
|
+-- Approval-specific
|   Approver User ID
|   Approval Timestamp
```



14. REDEMPTION METRICS FLOW

=====

[REDEMPTION ENQUIRY]



[RECORD ENQUIRY]



DECISION: CURRENT STATUS



+--> [PENDING]



+--> [AWAITING_SHORTFALL_RESOLUTION]



+--> [APPROVED] → [LEDGER TRANSACTION] → [BALANCE UPDATED]



+--> [REJECTED]



+--> [CANCELLED]



[REDEMPTION METRICS]



+--> [PENDING]



- COUNT



- VALUE



+--> [AWAITING_SHORTFALL_RESOLUTION]



- COUNT



- VALUE



+--> [APPROVED]



- COUNT



- VALUE



+--> [REJECTED]



- COUNT



- VALUE



+--> [CANCELLED]



- COUNT



- VALUE



[BY REDEMPTION TYPE]



[COURSE, REFUND, REINVEST, DONATION, ACCESSORIES/GADGETS]



[ADMIN DASHBOARD]



[REPORTS]

15. METRICS / DASHBOARD FLOW

=====

[PAYMENT / PLAN / REDEMPTION / REFERRAL EVENT]



[TRANSACTION / LEDGER DATA]



[METRICS AGGREGATION]



+--> [STUDENT METRICS]



↓
[STUDENT DASHBOARD]



+--> [ADMIN METRICS]



[ADMIN DASHBOARD]



[REPORTS]

STUDENT METRICS

- Total Invested
- Total Interest
- Redeemable Balance
- Reward Points
- Active Plans
- Closed Plans
- Next Deduction
- Referral Earnings
- Transaction Graph

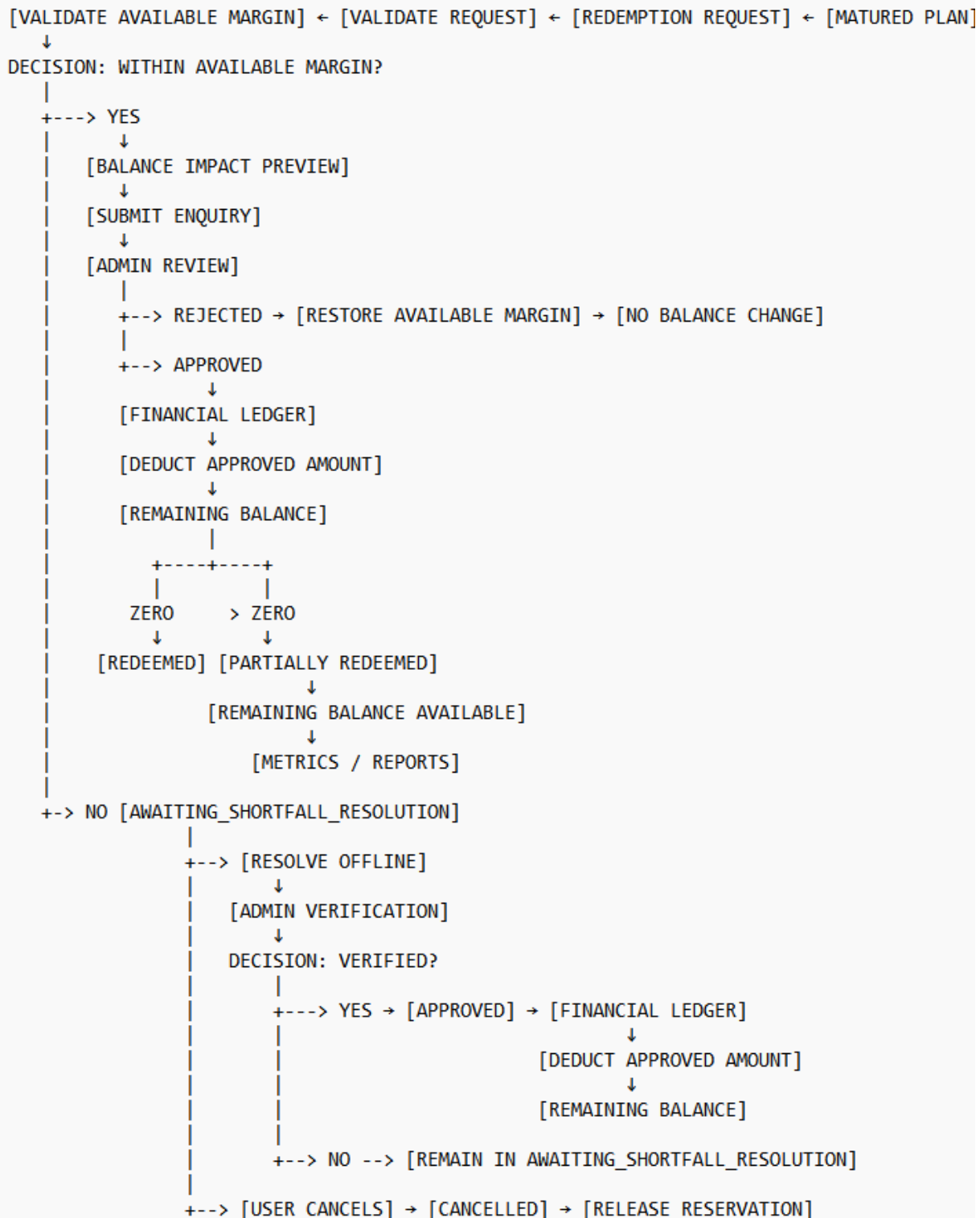
ADMIN METRICS

- Total Users
- Active Plans
- Today's Payments
- Pending Refunds
- Pending Enquiries
- Referral Payout
- Revenue
- Failed Payments
- Upcoming Maturities

DATA PRINCIPLE

Dashboard metrics should be derived from transactional and ledger data rather than independently maintaining conflicting financial values.

16. COMPLETE END-TO-END REDEMPTION FLOW



END-TO-END BUSINESS SEQUENCE :-

- Plan reaches maturity.
- User initiates an eligible redemption.
- System validates the request and Available Margin.
- If the request exceeds Available Margin, it enters AWAITING_SHORTFALL_RESOLUTION.
- The user resolves the shortfall offline or cancels the request.
- Admin verifies the offline resolution before approval.
- Only an approved request proceeds to financial processing.
- Ledger and balance are updated after approval.
- The resulting redemption status and metrics are recorded.

17. Commission Lifecycle

Commission Lifecycle:

ACCRUED → APPROVED → CREDITED → AVAILABLE FOR WITHDRAWAL → WITHDRAWN

Non-Accrual Outcome: Payment/Eligibility Evaluation → NOT_ACCRUED

NOT_ACCRUED shall be used only when an eligible commission opportunity is evaluated but no commission is generated, such as when:

- the underlying payment fails;
- the payment is skipped and no commission is earned;
- the payment does not satisfy the configured commission eligibility criteria; or
- the referral relationship is no longer eligible for future commission accrual.

NOT_ACCRUED is a **terminal non-financial state** and shall not represent a commission balance, payable amount, or ledger credit.

A NOT_ACCRUED record shall have a commission amount of ₹0 and shall not progress to APPROVED, CREDITED, AVAILABLE FOR WITHDRAWAL, or WITHDRAWN.

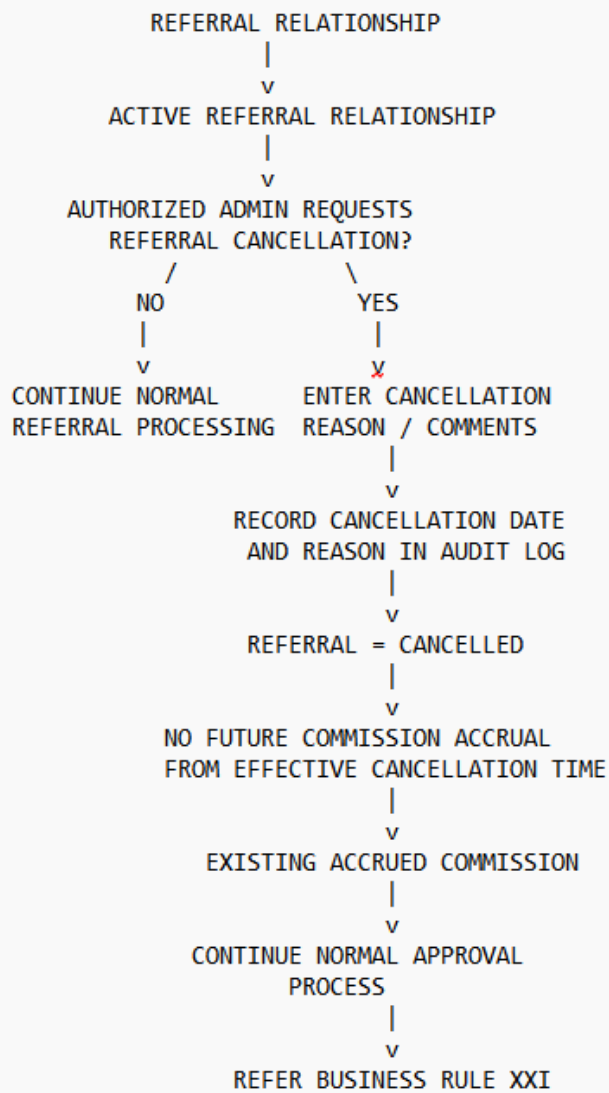
A failed payment shall therefore result in **no commission accrual** rather than a failed commission transaction.

Commission States:

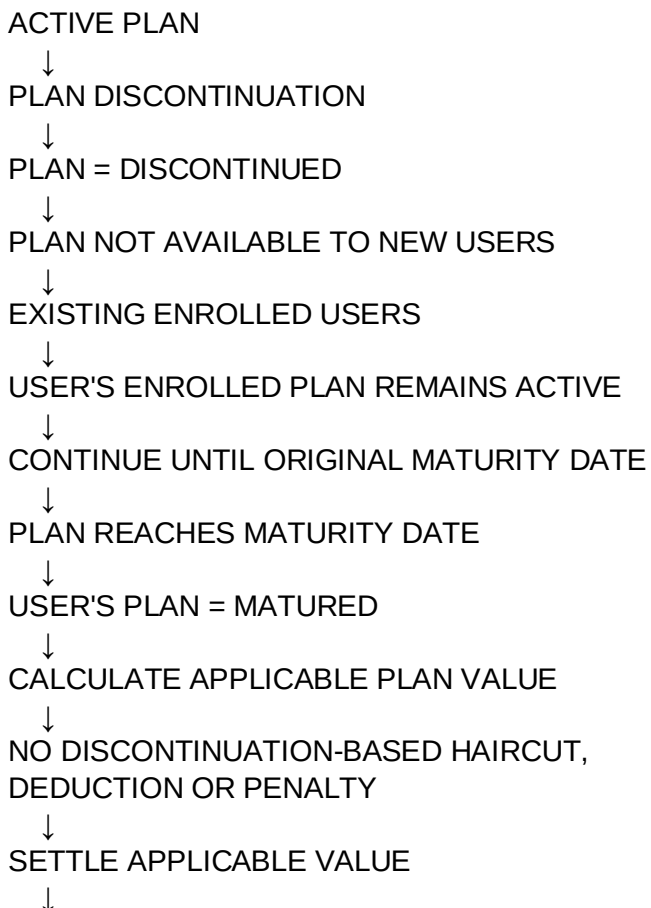
- **ACCRUED:** Commission has been calculated and recorded but has not yet been approved.
- **APPROVED:** Commission has passed the applicable business validation/approval stage.
- **CREDITED:** Commission has been added to the Referrer's commission balance.
- **AVAILABLE FOR WITHDRAWAL:** Commission is available for withdrawal by the Referrer.
- **WITHDRAWN:** Commission has been withdrawn by the Referrer.
- **NOT_ACCRUED:** It shall be used when a commission opportunity is evaluated but commission is not generated because one or more of the following conditions apply:
 - The payment is failed or unsuccessful.
 - The payment is not received within the applicable grace period.
 - The payment does not qualify under the configured commission type or commission cycle.
 - The Referrer is not eligible for commission for the applicable plan/payment.
 - The referral relationship is no longer eligible for future commission accrual.
 - The commission has already been generated where the applicable commission rule is One-Time Commission.

NOT_ACCRUED shall have a commission amount of ₹0 and shall not create a financial credit or ledger balance. It shall not progress to APPROVED, CREDITED, AVAILABLE FOR WITHDRAWAL, or WITHDRAWN.

18. Referral Cancellation:



19. Plan Discontinuation:



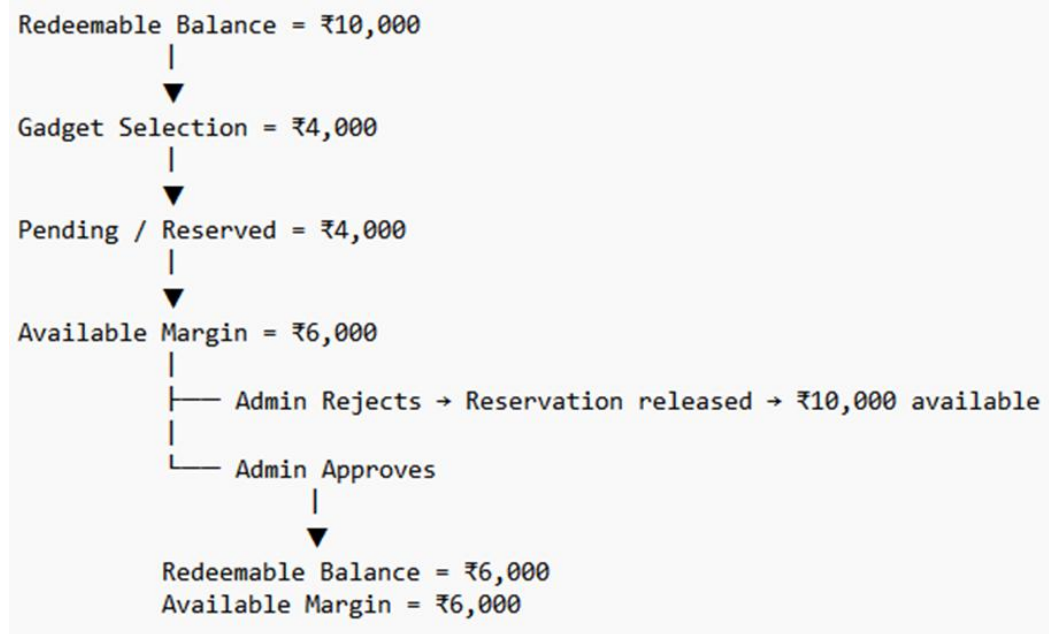
UPDATE BALANCE / LEDGER



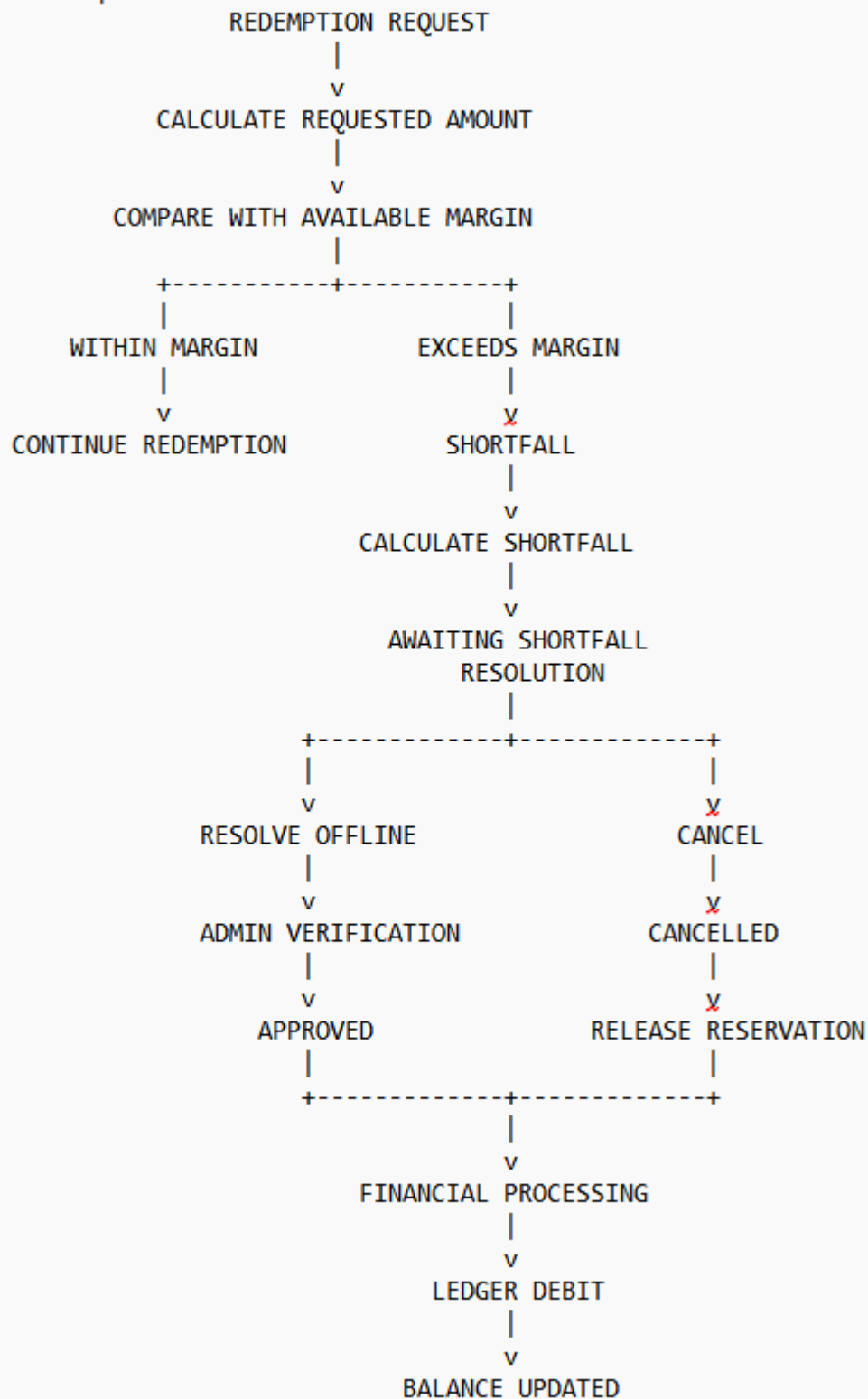
NORMAL REDEMPTION FLOW

- Plan status: DISCONTINUED — no longer available to new users.
- Existing user's enrolment: remains ACTIVE until its original maturity date.
- At maturity: the enrolled user's plan becomes MATURED.
- No early settlement merely because the plan was discontinued.

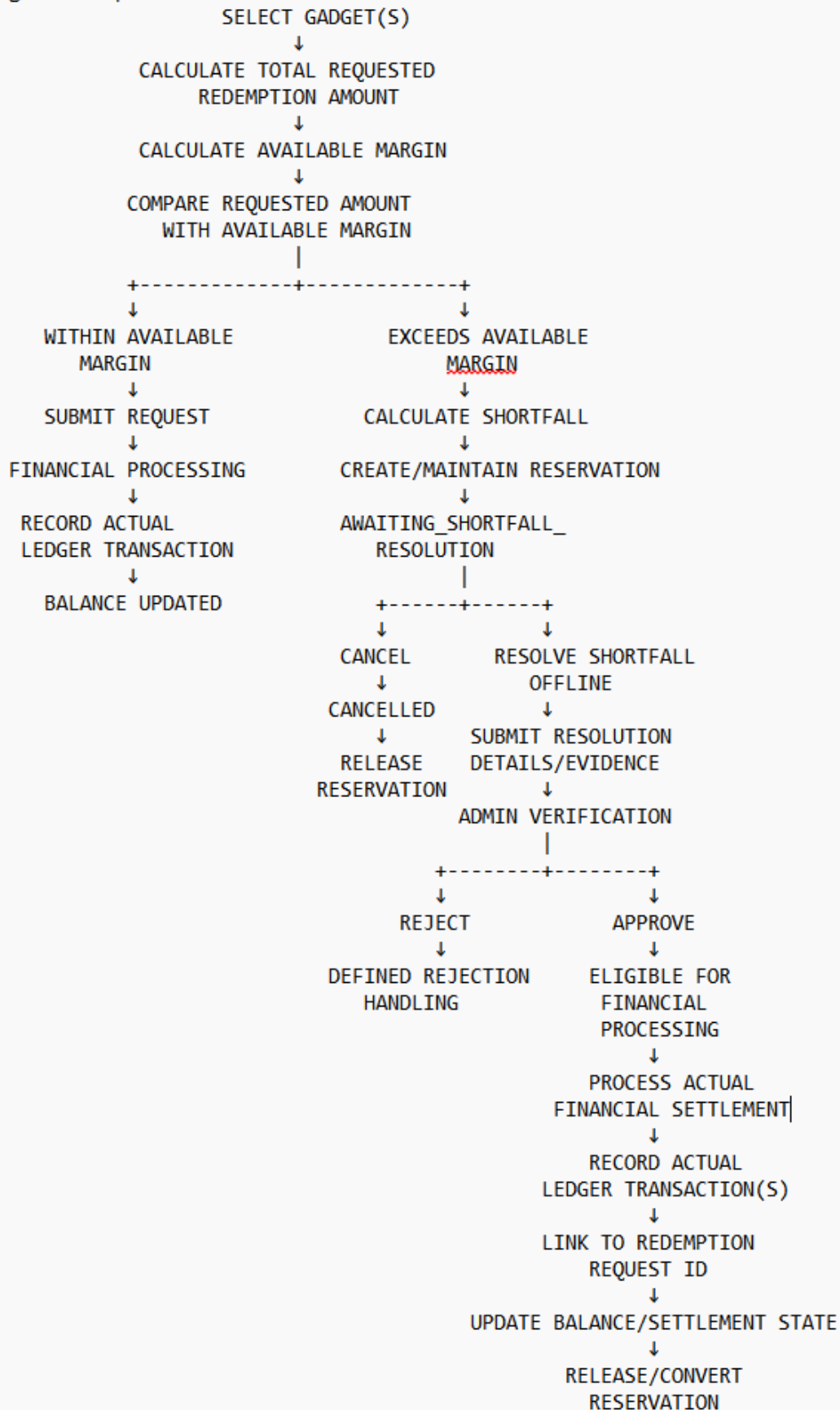
20. Gadget Redemption Calculation:-



21. Common Redemption Shortfall Flow:



22. Gadget Redemption - Shortfall



For the shortfall branch:

AWAITING_SHORTFALL_RESOLUTION → Offline Resolution → Admin Verification → Approval → Financial Processing → Ledger Transaction

There shall be **no ledger debit, no partial redemption, and no financial settlement between the shortfall identification and Admin approval stages.**

Gadget Redemption — Shortfall Scenario

A user selects gadgets with a total requested redemption value of **₹12,000**.

At the time of submitting the redemption request:

Parameter	Amount
Actual Redeemable Balance	₹10,000
Existing Active Reservations	₹0
Available Margin	₹10,000
Requested Redemption Amount	₹12,000
Shortfall Amount	₹2,000

Shortfall Calculation

Available Margin = Actual Redeemable Balance – Total Active Reserved/Committed Redemption Amount

Available Margin = ₹10,000 – ₹0 = ₹10,000

Shortfall = Requested Redemption Amount – Available Margin

Shortfall = ₹12,000 – ₹10,000 = ₹2,000

Since the requested redemption amount of **₹12,000** exceeds the Available Margin of **₹10,000**, the redemption request shall move to:

AWAITING_SHORTFALL_RESOLUTION

While the Shortfall Is Pending

The system shall:

- Display the Requested Redemption Amount as ₹12,000.
- Display the Available Margin as ₹10,000.
- Display the Shortfall Amount as ₹2,000.
- Keep the **entire redemption request financially unprocessed**.
- Not process the ₹10,000 Available Margin portion separately.
- Not create a redemption ledger debit.
- Not reduce the Redeemable Balance as a result of redemption settlement.
- Maintain any applicable reservation required by the redemption lifecycle.
- Require the user to resolve the ₹2,000 shortfall outside the system.

Therefore:

Redemption Ledger Debit = ₹0

Redemption Financial Settlement = ₹0

Offline Shortfall Resolution

The user resolves the **₹2,000 shortfall outside the system** through the prescribed offline process and submits the required details/evidence.

The system records the shortfall-resolution information against the same **Redemption Request ID**.

The offline resolution itself does **not** create a redemption ledger debit.

Admin Verification

Admin verifies the submitted shortfall resolution.

If Admin rejects the resolution, the redemption follows the applicable rejection/failure handling and no redemption ledger debit is created.

If Admin approves the resolution, the redemption becomes eligible for financial processing.

Admin Shortfall Verification SLA:

After the user submits the required shortfall-resolution details/evidence, the authorized Admin shall review and verify the submission within the configured shortfall verification period.

- **Default verification period:** 24 hours.
- **Configurable range:** 1 to 72 hours.
- The verification period shall start from the timestamp at which the user submits the required shortfall-resolution details/evidence.
- The system shall record the submission timestamp and verification timestamp.
- If verification is not completed within the configured verification period, the redemption request shall remain subject to the applicable system-wide Redemption Request Expiry rules.
- The system-wide Redemption Request Expiry shall take precedence if the redemption request reaches its expiry before Admin verification is completed.

Financial Processing After Approval

After Admin approval, the system processes the redemption using the **defined financial settlement mechanism**.

The system shall record the actual financial transaction(s) resulting from that settlement and link them to the same **Redemption Request ID**.

The system shall **not**:

- debit ₹10,000 first and ₹2,000 later as two independent redemption transactions;
- create a fictitious ₹12,000 debit against a ₹10,000 Redeemable Balance; or
- create a ledger debit merely to represent the offline shortfall.

The ledger shall record only the **actual financial transaction(s)** that occur as part of the approved settlement.

Cancellation

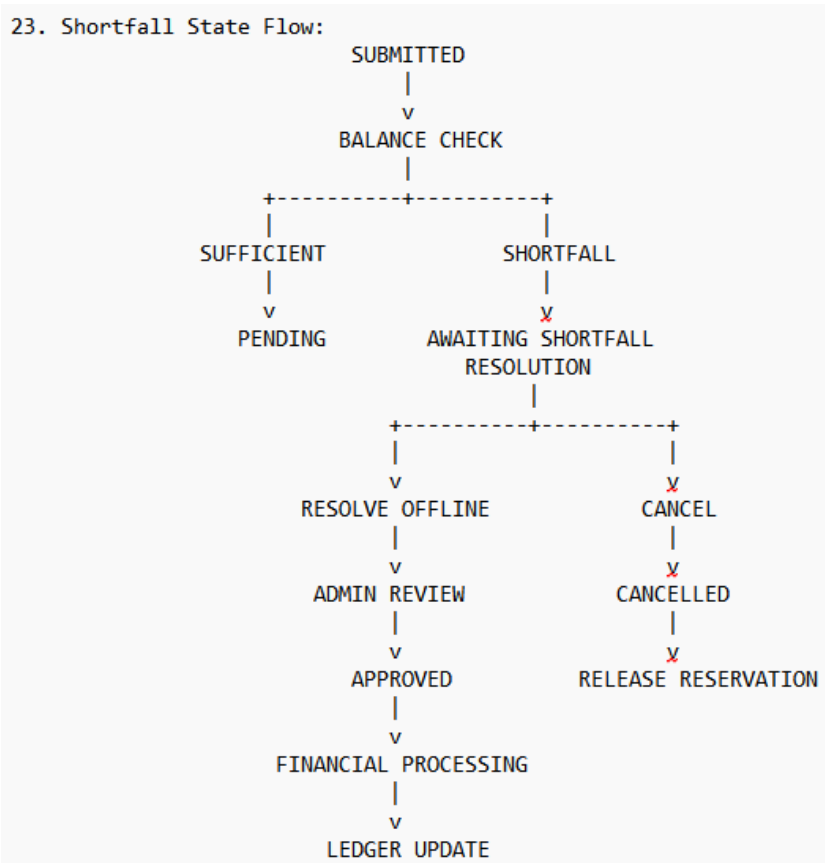
If the user cancels the redemption before financial settlement:

Item	Result
Redemption Status	CANCELLED
Financial Settlement	₹0
Redemption Ledger Debit	₹0
Reservation	Released

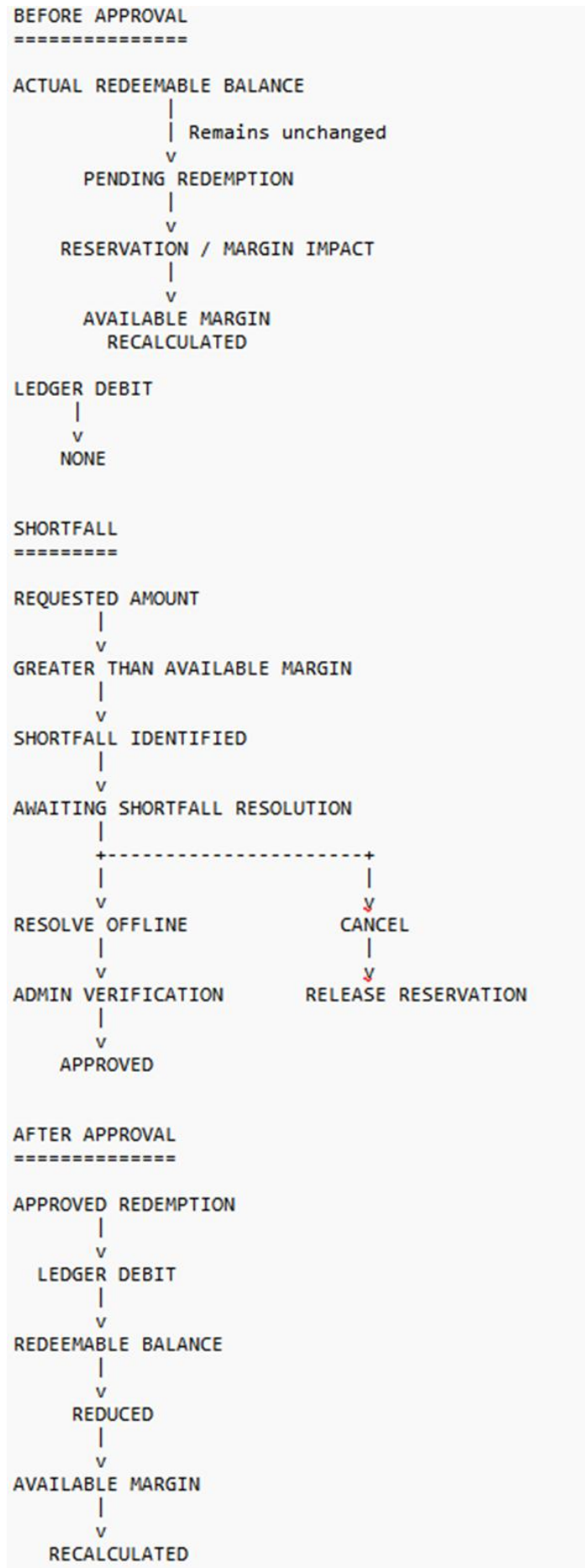
Important Accounting Principle

In this example, the **₹2,000 shortfall is a prerequisite for approving the ₹12,000 redemption request; it is not automatically a ₹2,000 ledger transaction.**

The redemption therefore remains **one redemption request**, with no partial online settlement while the shortfall is unresolved.



24. Financial Impact - Before and After Approval:-



After Approval:

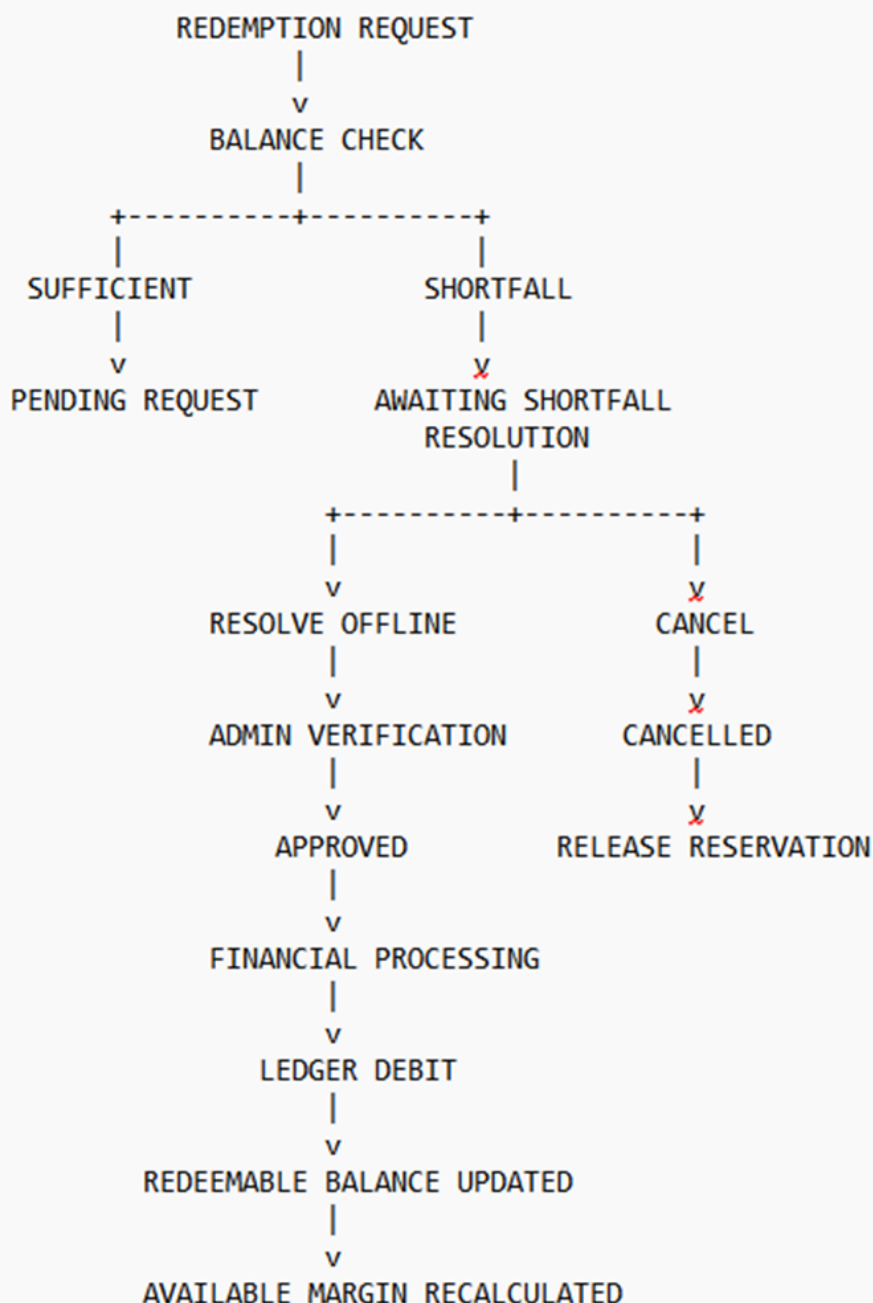
- The approved redemption amount shall be processed through the Unified Financial Ledger.
- The applicable redemption amount shall be debited from the user's Redeemable Balance.
- The Available Margin shall be recalculated after the balance update.

- The approval-to-financial-settlement flow shall apply consistently to Course, Refund, Reinvestment, Donation, Gadgets & Accessories, and Franchisee redemptions, subject to the specific settlement mechanics defined for each category.

Before Approval:

- The Actual Redeemable Balance shall remain unchanged.
- Any applicable redemption reservation shall affect only the Available Margin.
- No redemption ledger debit or financial settlement shall occur before Admin approval.
- For a shortfall request, the full redemption shall remain financially unprocessed until the shortfall is resolved and verified by Admin.
- If the request is cancelled, the associated reservation shall be released.

25. Balance and Ledger Impact:-

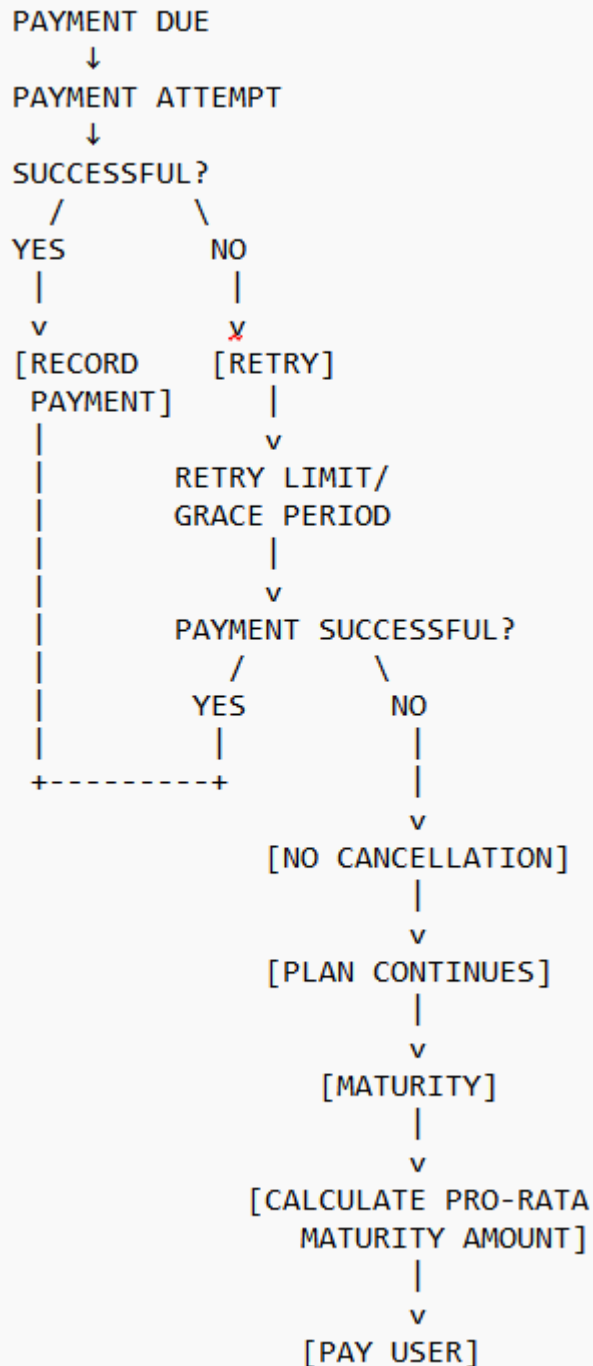


Balance and Ledger Impact:

- On submission, the system shall perform the applicable balance/Available Margin check.
- A sufficient-balance request shall proceed to PENDING and may create a reservation.
- A shortfall request shall move to AWAITING_SHORTFALL_RESOLUTION and shall not create a financial ledger debit.
- Resolution of an offline shortfall shall be recorded as verification information/evidence and shall not itself create a redemption ledger debit.

- After Admin approval, the approved redemption shall create the applicable Unified Financial Ledger entry and update the user's Redeemable Balance.
- Available Margin shall then be recalculated based on the updated balance and active reservations.
- On cancellation, rejection, or expiry, associated reservations shall be released without creating a redemption debit.

26. Failed Payment and Retry Management



27. Franchisee Redemption Flow:

ADMIN CREATES FRANCHISEE PLAN



ENTER PLAN NAME + ONE-TIME DEDUCTIBLE PRICE



MAP COLLEGES



USER OPENS REDEEM → FRANCHISEE



VIEW FRANCHISEE PLANS



SELECT PLAN



VIEW MAPPED COLLEGES



SELECT COLLEGE



DISPLAY ONE-TIME DEDUCTIBLE PRICE



CHECK AVAILABLE MARGIN



PRICE ≤ AVAILABLE MARGIN?

YES → Submit Enquiry → Admin Review → Approve → Financial Processing → Ledger Update → Balance Update

NO → Calculate Shortfall → Submit Enquiry → AWAITING_SHORTFALL_RESOLUTION → Offline Resolution → Admin Verification → Approval → Financial Processing → Ledger Update → Balance Update

Worked Financial Examples – Redemption and Ledger Calculations:

Illustrative examples only. Actual interest rates, reward percentages, and commission percentages are configurable by Admin.

1. Course Redemption

Scenario A: Redeeming amount for course for the first time.

Example:

Item	Amount
Redeemable Balance	₹20,000
Course Fee	₹15,000
Reward Points Balance	₹0
Reward Percentage	10%
Reward Points Earned	₹1,500
Balance After Approval	₹5,000
Reward Points Balance	₹1,500

Flow:

Redeemable Balance

₹20,000



Reward Points Balance

₹0



Course Redemption

₹15,000



Balance After Redemption

₹5,000

↓
 Reward = ₹15,000 × 10%
 ↓
 Reward Points Balance
 ₹1,500
 ↓
 Usable Only for Next Course

Scenario B: Redeeming amount for course for the second time.
 Example:

Item	Amount
Redeemable Balance	₹20,000
Course Fee	₹15,000
Reward Points Balance Before Redemption	₹1,500
Reward Percentage	10%
Redeemable Balance Used	₹13,500
Reward Points Used	₹1,500
Balance After Approval	₹6,500
Reward Points Earned	₹1,500

Flow:

Redeemable Balance
 ₹20,000
 ↓
 Reward Points Balance
 ₹1,500
 ↓
 Course Redemption
 ₹15,000 (₹13,500 Redeemable Balance + ₹1,500 Reward Points)
 ↓
 Redeemable Balance After Redemption
 ₹6,500
 ↓
 Reward = ₹15,000 × 10%
 ↓
 Reward Points Earned = ₹1,500

Note: Reward Points do not increase the redeemable balance and cannot be withdrawn or used for refund, donation, reinvestment, or gadgets.

Reward Points Rounding:

Reward Points shall be calculated using:

Reward Points = Applicable Course Fee × Configured Reward Percentage

Where the calculation produces a fractional value, the system shall **round down (floor) the calculated Reward Points to the nearest whole Reward Point.**

Example:

Applicable Course Fee = ₹12,345

Reward Percentage = 10%

Calculated Reward = ₹1,234.50

Reward Points Credited = 1,234

The system shall not round the Reward Points upward.

Any fractional amount discarded through rounding shall not be credited to the user's Reward Points balance.

Reward Points Balance and Redemption Rules:

- **Data Type:** Reward Points shall be stored as non-negative whole numbers. Fractional Reward Points shall not be stored or maintained.
- **Expiry:** Reward Points shall **not expire** unless an expiry period is subsequently configured by the Admin.
- **Maximum Balance:** No maximum cap shall apply to the user's Reward Points balance unless a cap is subsequently configured by the Admin.
- **Redemption:** When Reward Points are used for an eligible redemption, the system shall deduct only the number of Reward Points actually used and retain the remaining balance.
- **Partial Redemption:** Partial use of Reward Points shall be permitted. The system shall calculate and record the exact whole-number Reward Points deducted and the resulting remaining Reward Points balance.
- **Insufficient Points:** A user shall not be permitted to redeem more Reward Points than the available Reward Points balance.
- **No Negative Balance:** The Reward Points balance shall never become negative.

2. Refund — Partial

Example:

Redeemable Balance	₹20,000
Requested Refund	₹8,000
Available Margin	₹12,000

Before Admin Approval:

Redeemable Balance	₹20,000
Available Margin	₹12,000

After Approval:

Redeemable Balance	₹12,000
Available Margin	₹12,000

3. Reinvestment — Partial

Example:

Redeemable Balance	₹30,000
Reinvestment Amount	₹20,000
Remaining Balance	₹10,000

Old Plan

```

|
+---- ₹20,000 ---> New Plan
|
+---- ₹10,000 ---> Remaining Redeemable Balance

```

Reinvestment is optional; the remaining ₹10,000 does not have to be reinvested.

4. Donation — Partial

Example:

Redeemable Balance	₹25,000
Donation Requested	₹5,000
Remaining Balance	₹20,000

Before approval:

Actual Balance	₹25,000
Available Margin	₹20,000

After approval:

Actual Balance	₹20,000
Available Margin	₹20,000

After Approval:

Approved Donation ₹5,000
 Unified Financial Ledger → Redemption/DONATION Debit ₹5,000
 Redeemable Balance Before ₹25,000
 Redeemable Balance After ₹20,000
 Available Margin recalculated based on the updated balance and active reservations.

Before Approval: No financial ledger debit and no reduction of Actual Redeemable Balance.

5. Gadgets & Accessories — Multiple Items

Initial State:

Description	Amount
Redeemable Balance	₹10,000
Available Margin	₹10,000

User selects gadgets:

Description	Amount
Selected Gadget A	₹2,500
Selected Gadget B	₹1,500
Total Requested	₹4,000

Before Admin Approval:

Description	Amount
Actual Redeemable Balance	₹10,000
Pending/Reserved Redemption	₹4,000
Available Margin	₹6,000

The ₹4,000 redemption amount is pending approval and is therefore reserved. The **Actual Redeemable Balance remains ₹10,000 until the redemption is approved.**

The Available Margin is calculated as:

$$\text{₹10,000} - \text{₹4,000} = \text{₹6,000}$$

Scenario A — User Adds a ₹7,000 Gadget After the Original Redemption Is Approved

The original ₹4,000 redemption has already been approved and processed.

Description	Amount
Current Actual Redeemable Balance	₹6,000
Additional Gadget	₹7,000
Shortfall	₹1,000

Calculation: $\text{₹7,000} - \text{₹6,000} = \text{₹1,000}$ shortfall

Therefore:

Description	Amount
Available Margin	₹6,000
Requested Additional Amount	₹7,000
Shortfall	₹1,000

The additional redemption cannot be approved while the ₹1,000 shortfall remains unresolved. The user must resolve the shortfall offline or cancel the redemption request.

Scenario B — User Adds a ₹7,000 Gadget Before the Original Redemption Is Approved

In this scenario, the original ₹4,000 redemption is **still pending approval**. Therefore, the ₹4,000 has not yet been debited from the Actual Redeemable Balance.

The user modifies the pending gadget cart by adding another ₹7,000 gadget.

Description	Amount
Existing Selection	₹4,000
Additional Gadget	₹7,000
Revised Total Requested	₹11,000
Actual Redeemable Balance	₹10,000
Shortfall	₹1,000

Calculation:

₹4,000 + ₹7,000 = ₹11,000

₹11,000 – ₹10,000 = ₹1,000 (shortfall)

Revised Pending State:

Description	Amount
Actual Redeemable Balance	₹10,000
Revised Pending Redemption Amount	₹11,000
Available Margin	₹0
Shortfall	₹1,000
Ledger Debit	₹0

The original ₹4,000 pending reservation is **replaced by the revised ₹11,000 pending request**. It must not be added to the revised request again.

Because the revised request exceeds the Actual Redeemable Balance by ₹1,000, the request cannot be approved while the shortfall remains unresolved.

The user must either:

1. **Resolve the ₹1,000 shortfall offline**, after which the Admin may verify and approve the request; or
2. **Cancel the redemption**, in which case no ledger debit is made.

When a pending gadget redemption cart is modified, the previous pending request shall be replaced by the revised total requested amount. The system shall recalculate the Available Margin and Shortfall using the revised request and shall not double-count the previous pending reservation.

When a pending redemption request is modified, the revised total requested amount shall replace the previous pending request amount for the purpose of calculating Available Margin and Shortfall. The previous pending amount shall not be counted separately.

Depiction of example as flow diagram:

INITIAL

Actual Redeemable Balance = ₹10,000

Available Margin = ₹10,000

|

v

USER SELECTS A + B

Requested Amount = ₹4,000

|

v

PENDING APPROVAL

Actual Redeemable Balance = ₹10,000

Pending Amount = ₹4,000

Available Margin = ₹6,000

Ledger Debit = ₹0

|

+-----+-----+

|

|

v

v

APPROVED

EDIT CART

|

|

v

v

Balance = ₹6,000 Revised Request = ₹11,000

Ledger = ₹4,000 Balance = ₹10,000

|

Shortfall = ₹1,000

|

|

v

v

NEW ₹7,000

RESOLVE SHORTFALL

REQUEST

OR CANCEL

|

v

SHORTFALL = ₹1,000

Full vs Partial Redemption:

Partial

Balance Before ₹50,000

Approved Redemption ₹15,000

Balance After ₹35,000

Plan Status: MATURED → PARTIALLY REDEEMED

Full

Balance Before ₹50,000

Approved Redemption ₹50,000

Balance After ₹0

Plan Status: MATURED → REDEEMED

This will make PARTIALLY_REDEEMED state mathematically unambiguous.

Partial Redemption Shortfall Example

REDEEMABLE BALANCE: INR 50,000

REQUESTED REDEMPTION AMOUNT: INR 70,000

AVAILABLE MARGIN: INR 50,000

SHORTFALL: INR 20,000

Shortfall Calculation:

Shortfall = Requested Redemption Amount – Available Margin = ₹70,000 – ₹50,000 = ₹20,000

Before Shortfall Resolution:

Redemption Ledger Debit: ₹0

Redeemable Balance: ₹50,000

After the user resolves the ₹20,000 shortfall offline and Admin approves the resolution, the redemption becomes eligible for financial processing in accordance with the applicable settlement mechanism.

Redeemable Amount Available for System-Side Settlement: ₹50,000

Redeemable Balance After System-Side Settlement: ₹0

Offline Shortfall Resolution: ₹20,000

Total Redemption Value: ₹70,000

Course Redemption example:

If:

Redeemable Balance ₹1,00,000

Course Fee ₹80,000

Shortfall ₹0

Then:

Redeemable Balance Deduction ₹80,000

Offline Payment ₹0

Remaining Balance ₹20,000

Ledger Debit ₹80,000

Course Redemption Shortfall Example:

Suppose:

Course Fee ₹80,000

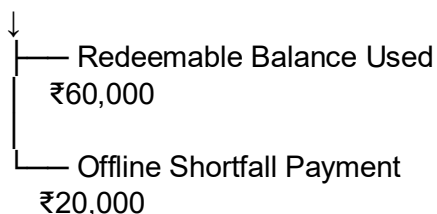
Redeemable Balance ₹60,000

Shortfall ₹20,000

The financial treatment is:

COURSE FEE

₹80,000



After successful offline payment and Admin approval:

Initial Redeemable Balance: ₹60,000

Redeemable Amount Used: -₹60,000

Remaining Redeemable Balance: ₹0

Offline Shortfall Payment ₹20,000

Total Course Fee ₹80,000

One Illustrative Interest Example:

Illustrative only — not a fixed business rate.

For example:

Principal ₹10,000

Illustrative annual rate 12%

Tenure 1 year

Interest method Simple Interest

Interest = ₹10,000 × 12% × 1 = ₹1,200

Illustrative maturity amount = ₹11,200

Successful Four Consecutive Payments + Referral Commission:

Configured recurring commission = ₹250 per completed four-payment cycle.

1st payment → Successful → Commission cycle: 1/4 → NOT_ACCRUED

2nd payment → Successful → Commission cycle: 2/4 → NOT_ACCRUED

3rd payment → Successful → Commission cycle: 3/4 → NOT_ACCRUED

4th payment → Successful → Commission cycle completed: 4/4 → ₹250 commission → ACCRUED

5th payment → Successful → New commission cycle: 1/4 → NOT_ACCRUED

6th payment → Successful → New commission cycle: 2/4 → NOT_ACCRUED

After the fourth consecutive successful payment, the ₹250 recurring commission is generated and the commission cycle restarts from 1/4 for subsequent payments.

Failed Payment + Referral Commission:

1st payment → Successful → Commission cycle: 1/4 → NOT_ACCRUED

2nd payment → Successful → Commission cycle: 2/4 → NOT_ACCRUED

3rd payment → Failed → ₹0 commission → NOT_ACCRUED

3rd payment → Later successful → Commission cycle: 3/4 → NOT_ACCRUED

4th payment → Successful → Commission cycle completes: 4/4 → ₹250 commission → ACCRUED

Illustrative example: Configured recurring commission = ₹250 per completed four-payment cycle.

A failed payment shall not count toward the four-payment commission cycle. If the failed payment is subsequently received successfully within the configured grace period, it shall count as a successful payment toward the same four-payment cycle. Once four consecutive successful payments are completed, the applicable recurring commission shall be generated and the commission cycle shall restart.

Franchisee Example

Normal case

One-Time Deductible Price = ₹40,000

Available Margin = ₹50,000

₹40,000 ≤ ₹50,000, so the User submits the Franchisee enquiry normally. Before approval, the Redeemable Balance remains unchanged, consistent with the existing enquiry model.

Shortfall case

One-Time Deductible Price = ₹80,000

Available Margin = ₹50,000

Shortfall = ₹80,000 – ₹50,000 = **₹30,000**

Before Admin approval:

Redeemable Balance remains unchanged.

Redemption Ledger Debit = ₹0.

After offline shortfall resolution and Admin approval:

System-side settlement shall be processed only in accordance with the applicable redemption settlement rules.

The ₹30,000 offline shortfall shall remain

Definition of Commission reporting:

Admin shall be able to view referral commission reports by:

- Referrer
- Referred User
- Plan
- Payment/transaction
- Commission amount
- Commission status
- Accrual date
- Approval date

- Credit date
- Withdrawal date
- Referral status

Final Referral Rules:

Self-Referral Prevention: The system shall prevent a user from registering or being associated as a Referred User using their own referral code. The system shall reject any referral relationship where the **Referrer User ID is the same as the Referred User ID**.

{Referrer User ID == Referred User ID} → REJECT REFERRAL

Single-Level Referral: The system shall support only direct Referrer → Referred User relationships. A Referrer may have one or more directly Referred Users. A Referred User may subsequently become a Referrer and refer newly registering users using their own referral code. Such subsequent referral relationships shall remain independent direct relationships and shall not create any multi-level, hierarchical, or indirect commission entitlement.

Direct Commission Eligibility: A Referrer shall receive commission only from eligible payments made by their directly Referred Users. Commission generated from a downstream referral shall be payable only to the Referrer directly associated with that downstream Referred User. No upstream Referrer shall receive commission from downstream or indirect referrals.

Referral Validity Period Configuration: The Admin shall configure a **global referral validity period**, expressed in days, applicable to newly created referral relationships.

Configuration Changes: Changes to the global referral validity period shall apply only to referral relationships created after the configuration change. Existing referral relationships shall retain their originally calculated expiry date.

Referral Cancellation: Refer business rule XX.

Existing Commission: Commission that was eligible and accrued before Referral Validity Period or cancellation shall continue through the normal commission approval process.

Referral Code Rotation: Only the active referral code shall be accepted for new referral registrations. When a code is regenerated, the old code shall immediately become inactive for new registrations. Code regeneration shall not alter, cancel, or reassign any existing Referrer → Referred User relationship or previously earned commission.

Referral Code Audit: The system shall record referral-code generation, regeneration, activation/deactivation timestamp, and the user responsible for the action in the audit log.

Referral relationships:

A → B

A → C

B → D

Commission relationships:

B payment



Commission → A

C payment



Commission → A

D payment



Commission → B

There is no:-

D payment



Commission → A

Commission States:

ACCRUED: Commission has been calculated but not yet approved.

APPROVED: Commission has passed the business validation/approval stage.
CREDITED: Commission has been added to the referrer's commission balance.
AVAILABLE FOR WITHDRAWAL: Referrer can request withdrawal.
WITHDRAWN: Commission has been withdrawn by the referrer.

Summary metrics:

Total Referral Commission Accrued
Total Commission Approved
Total Commission Credited
Total Commission Withdrawn
Total Pending Commission

Security Configuration / Technical Security Requirements:

Control	Requirement
Password Hashing	Passwords shall be stored only as securely salted password hashes using Bcrypt . Plain-text or reversibly encrypted passwords shall not be stored.
Password Complexity	Minimum 8 characters including at least one uppercase letter, one lowercase letter, one number, and one special character.
2FA	Two-factor authentication shall use OTP delivered through the registered email address for applicable authentication flows.
2FA OTP	OTP shall be 6 digits , valid for 5 minutes , and limited to 5 verification attempts per OTP. A maximum of 3 OTP resend requests within 15 minutes shall be permitted.
JWT Access Token	Access token shall expire after 30 minutes .
Refresh Token	Refresh token shall expire after 30 days and shall be invalidated on logout, password change, or account security reset.
Device / Session Tracking	The system shall track authenticated device/session information, including login time, logout time, and session status.
Login History	The system shall record successful and failed login attempts with timestamp and relevant security/audit information.
Failed Login Protection	After 5 consecutive failed login attempts , further login attempts for the account shall be temporarily blocked for 15 minutes .
Audit Logging	Security-sensitive events, including login success/failure, logout, password change/reset, 2FA events, token/session events, and account lockout shall be audit logged.
Password Reset	Password-reset tokens shall be single-use and shall expire after 30 minutes .
Transport Security	Authentication credentials, tokens, OTPs, and other sensitive information shall be transmitted only over HTTPS/TLS .
Social Login / OAuth:	Social authentication shall use OAuth 2.0 and/or OpenID Connect authorization flows with administrator-approved providers. The system shall validate the provider-issued authorization result/token, verify the identity claims required for account login, and reject invalid, expired, or unverifiable authentication responses. Provider credentials/secrets shall be stored securely and shall not be exposed to users.

JWT Access Token

Access tokens shall use **RS256 (RSA SHA-256)** signing. Access tokens shall expire after **30 minutes**. The signing private key shall be securely stored and shall not be exposed to clients.

Refresh Token

Refresh tokens shall expire after **30 days** and shall be invalidated on logout, password change, or account security reset. Refresh tokens shall be **single-use and rotated upon successful refresh**. The previously used refresh token shall be invalidated immediately after rotation. Reuse of an invalidated refresh token shall be rejected and the associated session shall be revoked.

Social Login and 2FA: Where 2FA is enabled for the user/account, successful social authentication shall not bypass the applicable 2FA requirement. The system shall initiate the configured email OTP 2FA step after successful social authentication. Users authenticating through social login shall use the registered email address for applicable 2FA, consistent with the existing 2FA rules.

Account Linking: If a social-login provider returns an email address that matches an existing verified user account, the system shall link the provider identity to the existing account only after the applicable account-verification rules are satisfied. The system shall not create a second account for the same verified email address.